

JAVA

ASSIGNMENT

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TO:-Mr. AMIT KUMAR

```
err eclipse-workspace - JavaCode\src\Termwork\MMANN.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JavaCode
package-info package-info pluppemen... mainT01.java araman.java sphere.java AreaAnimal... ThrowExample... MMANN.java
1 package Termwork;
2
3 import java.util.*;
4 public class MMANN {
5
6
7     public static void main(String[] args)
8     {
9         Scanner sc=new Scanner(System.in);
10         int x=sc.nextInt();
11         int sum =0;
12         while(x>0)
13         {
14             sum=sum+x;
15             x=sc.nextInt();
16         }
17         System.out.println(sum);
18     }
19 }
20 }
21 }
```

```
err eclipse-workspace - JavaCode\src\IAH\areaman.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JavaCode
package-info package-info pluppemen... mainT01.java areaman.java sphere.java AreaAnimal... ThrowExample... MMANN.java
1 package IAH;
2 import java.util.*;
3 class area
4 {
5     void area(int l,int b)
6     {
7         System.out.println("The area of rect is"+l*b);
8     }
9     void area(int r)
10    {
11        System.out.println("The area of circle is"+(3.14*r*r));
12    }
13 }
14
15 public class areamain {
16
17     public static void main(String[] args) {
18         int r,l,b;
19         Scanner sc=new Scanner (System.in);
20         System.out.println("Enter the value of r");
21         long r=sc.nextInt();
22         System.out.println("Enter the value of l");
23         l=sc.nextInt();
24         System.out.println("Enter the value of b");
25         b=sc.nextInt();
26         area a=new area();
27         a.area(l,b);
28         a.area(r);
29     }
30 }
31 }
32 }
```

```
My eclipse-workspace - JavaCode\src\LAB\sphere.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JavaCode
package-info package-info.java sphere.java
1 package-info;
2
3 import java.util.Scanner;
4
5 class solo
6 {
7     void sphereon(int r)
8     {
9
10    }
11    void sphereon(float r)
12    {
13
14    }
15 }
16 public class sphere {
17
18     public static void main(String[] args)
19     {
20         int r;
21         Scanner sc=new Scanner (System.in);
22         System.out.println("Enter the value of r");
23         r=sc.nextInt();
24         solo s=new solo();
25
26     }
27 }
28
29
30
```

```
My eclipse-workspace - JavaCode\src\LAB\AccessAnimal.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JavaCode
21 System.out.println(getName()+" is eating food");
22 }
23 }
24
25 class Dog extends Animal {
26     public Dog(String name) {
27         super(name); // Call the constructor of the Animal class
28     }
29
30     // implementing the abstract method for Dog
31     @Override
32     public void makeSound() {
33         System.out.println(getName() + " says: Bark");
34     }
35 }
36
37 class Cat extends Animal {
38     public Cat(String name) {
39         super(name); // Call the constructor of the Animal class
40     }
41
42     // implementing the abstract method for Cat
43     @Override
44     public void makeSound() {
45         System.out.println(getName() + " says: Meow");
46     }
47 }
48
49 public class AccessAnimal {
50
51     public static void main(String[] args) {
52         Animal dog = new Dog("Buddy");
53         dog.makeSound(); // Output: Buddy says: Bark
54         dog.eat(); //Output: Buddy is eating food
55
56         Animal cat = new Cat("Whiskers");
57         cat.makeSound(); // Output: Whiskers says: Meow
58         cat.eat(); //Output: Buddy is eating food
59     }
60 }

```

```
1 package IAH;
2
3 abstract class Animal {
4     // Field for the name of the animal
5     private String name;
6
7     // Constructor to set the name of the animal
8     public Animal(String name) {
9         this.name = name;
10    }
11
12    // Abstract method makeSound
13    public abstract void makeSound();
14
15    // Concrete method to get the name of the animal
16    public String getName() {
17        return name;
18    }
19
20    public void eat() {
21        System.out.println(getName()+" is eating food");
22    }
23 }
24
25 class Dog extends Animal {
26     public Dog(String name) {
27         super(name); // Call the constructor of the Animal class
28     }
29
30    // Implementing the abstract method for Dog
31    @Override
32    public void makeSound() {
33        System.out.println(getName() + " says: Bark");
34    }
35 }
36
37 class Cat extends Animal {
38     public Cat(String name) {
39         super(name); // Call the constructor of the Animal class
40    }
41 }
```

```
1 package Termwork;
2 import java.util.ArrayList;
3
4 public class AlternateElements {
5
6     public static List<Integer> alternate(List<Integer> list1, List<Integer> list2) {
7         List<Integer> result = new ArrayList<>();
8         int size1 = list1.size();
9         int size2 = list2.size();
10        int maxSize = Math.max(size1, size2);
11
12        for (int i = 0; i < maxSize; i++) {
13            if (i < size1)
14                result.add(list1.get(i));
15            if (i < size2)
16                result.add(list2.get(i));
17        }
18
19        return result;
20    }
21
22    public static void main(String[] args) {
23        List<Integer> list1 = new ArrayList<>();
24        list1.add(1);
25        list1.add(2);
26        list1.add(3);
27        list1.add(4);
28        list1.add(5);
29
30        List<Integer> list2 = new ArrayList<>();
31        list2.add(6);
32        list2.add(7);
33        list2.add(8);
34        list2.add(9);
35        list2.add(10);
36        list2.add(11);
37        list2.add(12);
38
39        List<Integer> result = alternate(list1, list2);
40        System.out.println(result);
41    }
42 }
```

```
My eclipse workspace - JavaCode\src\multiple\ThrowExample.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JavaCode
package multiple;

public class ThrowExample {
    static void checkAge(int age) throws Exception {
        if (age < 18) {
            // Throw an ArithmeticException if the age is less than 18
            throw new Exception("Access denied - "
                + "You must be at least 18 years old.");
        } else {
            System.out.println("Access granted - You are old enough!");
        }
    }

    public static void main(String[] args) {
        try {
            checkAge(10); // This will throw an ArithmeticException
        } catch (Exception e) {
            System.out.println("Exception caught in main: " + e);
        }
    }
}
```

```
My eclipse workspace - JavaCode\src\Termwork\ArrayDemo.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JavaCode
package Termwork;

import java.util.Arrays;

class ArrayDemo {

    // Overloaded arrayFunc method to find pairs with a specific sum
    void arrayFunc(int[] array, int target) {
        System.out.println("Pairs of elements whose sum is " + target + " are:");
        for (int i = 0; i < array.length; i++) {
            for (int j = i + 1; j < array.length; j++) {
                if (array[i] + array[j] == target) {
                    System.out.println(array[i] + " + " + array[j] + " = " + target);
                }
            }
        }
    }

    void arrayFunc(int[] A, int p, int[] B, int q) {
        int[] merged = new int[p + q];
        int i = 0, j = 0, k = 0;

        // Merge the two arrays
        while (i < p && j < q) {
            if (A[i] < B[j]) {
                merged[k++] = A[i++];
            } else {
                merged[k++] = B[j++];
            }
        }

        // Copy remaining elements from A
        while (i < p) {
            merged[k++] = A[i++];
        }

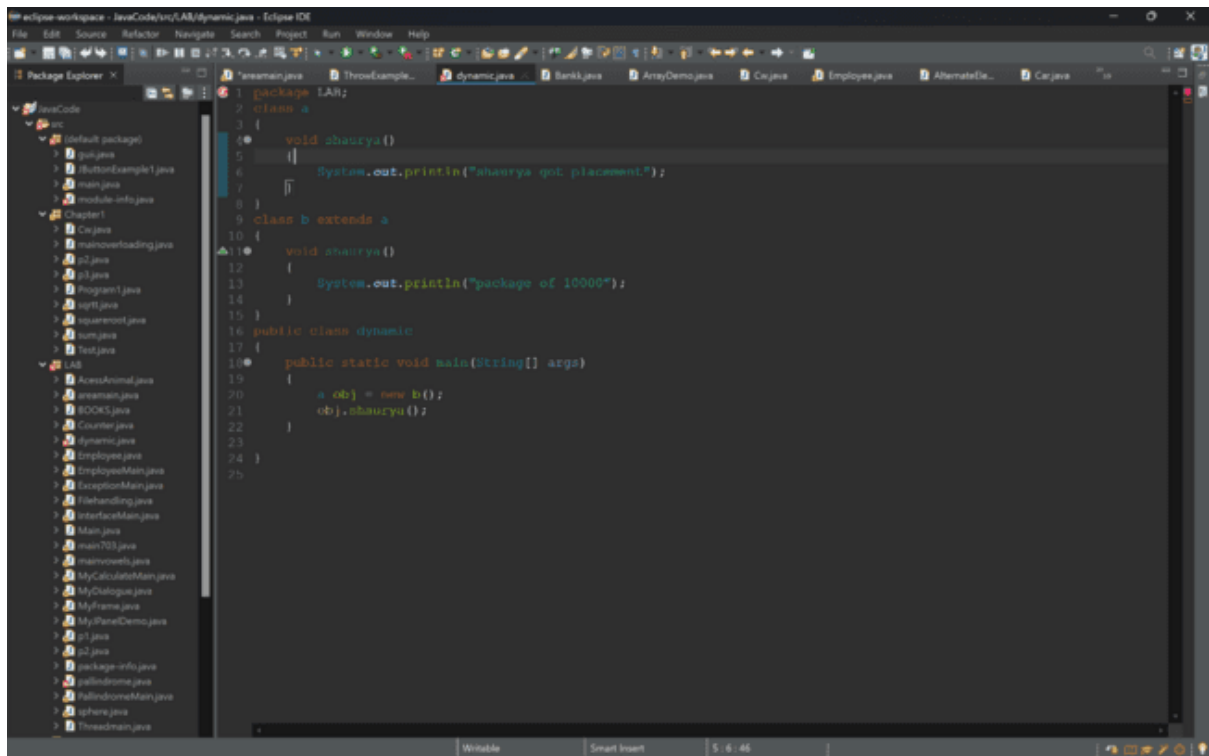
        // Copy remaining elements from B
        while (j < q) {
            merged[k++] = B[j++];
        }
    }
}
```

```
106 for (Bank account : bankAccounts) {
107     if (account.getAccountNumber() == accNumberToDisplay) {
108         account.displayInfo();
109     }
110 }
111 break;
112 case 2:
113     System.out.print("Enter account number: ");
114     int accNumberToDeposit = scanner.nextInt();
115     System.out.print("Enter amount to deposit: ");
116     double depositAmount = scanner.nextDouble();
117     scanner.nextLine(); // Consume newline
118     for (Bank account : bankAccounts) {
119         if (account.getAccountNumber() == accNumberToDeposit) {
120             account.deposit(depositAmount);
121             account.displayInfo();
122         }
123     }
124     break;
125 case 3:
126     System.out.print("Enter account number: ");
127     int accNumberToWithdraw = scanner.nextInt();
128     System.out.print("Enter amount to withdraw: ");
129     double withdrawAmount = scanner.nextDouble();
130     scanner.nextLine(); // Consume newline
131     for (Bank account : bankAccounts) {
132         if (account.getAccountNumber() == accNumberToWithdraw) {
133             account.withdraw(withdrawAmount);
134             account.displayInfo();
135         }
136     }
137     break;
138 case 4:
139     System.out.print("Enter account number: ");
140     int accNumberToChangeAddress = scanner.nextInt();
141     scanner.nextLine(); // Consume newline
142     System.out.print("Enter new address: ");
143     String newAddress = scanner.nextLine();
144     for (Bank account : bankAccounts) {
145         if (account.getAccountNumber() == accNumberToChangeAddress) {
```

```
122     }
123 }
124 break;
125 case 3:
126     System.out.print("Enter account number: ");
127     int accNumberToWithdraw = scanner.nextInt();
128     System.out.print("Enter amount to withdraw: ");
129     double withdrawAmount = scanner.nextDouble();
130     scanner.nextLine(); // Consume newline
131     for (Bank account : bankAccounts) {
132         if (account.getAccountNumber() == accNumberToWithdraw) {
133             account.withdraw(withdrawAmount);
134             account.displayInfo();
135         }
136     }
137     break;
138 case 4:
139     System.out.print("Enter account number: ");
140     int accNumberToChangeAddress = scanner.nextInt();
141     scanner.nextLine(); // Consume newline
142     System.out.print("Enter new address: ");
143     String newAddress = scanner.nextLine();
144     for (Bank account : bankAccounts) {
145         if (account.getAccountNumber() == accNumberToChangeAddress) {
146             account.changeAddress(newAddress);
147             account.displayInfo();
148         }
149     }
150     break;
151 case 5:
152     continueOperations = false;
153     break;
154 default:
155     System.out.println("Invalid choice, please try again.");
156 }
157 scanner.close();
158 }
159 }
160 }
161 }
```

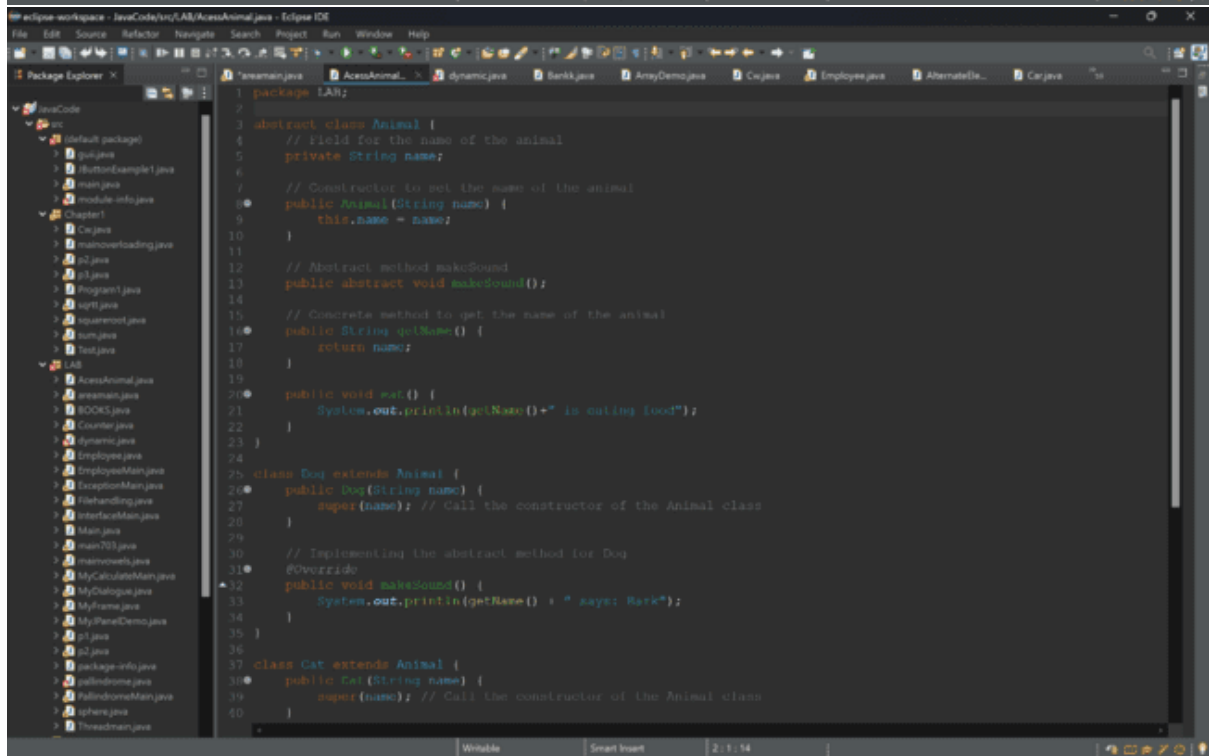
```
My eclipse-workspace - JavaCode\src\IAM\Employee.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JavaCode
IAM
1 package IAM;
2 /** Design a class named Employee with private fields for name, ID, and salary. Provide public setter and getter methods. */
3 import java.util.*;
4
5 public class Employee
6 {
7     private int ID;
8     private String Name;
9     private int Salary;
10    public Employee(int id,String N,int salary)
11    {
12        ID=id;
13        Name=N;
14        Salary=salary;
15    }
16
17    void disp()
18    {
19        System.out.println("ID="+ID+" Name="+Name+" Salary "+Salary);
20    }
21
22    class MAINEmployee
23    {
24        public static void main(String[] args)
25        {
26            Employee e1=new Employee(1234,"Shaurya Pandey",80000);
27            e1.disp();
28        }
29    }
30 }
31
```

```
My eclipse-workspace - JavaCode\src\IAM\aremain.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JavaCode
IAM
1 package IAM;
2 import java.util.*;
3
4 class area
5 {
6     void area(int l,int b)
7     {
8         System.out.println("The area of rect is:"+l*b);
9     }
10    void area(int r)
11    {
12        System.out.println("The area of circle is:"+3.14*r*r);
13    }
14 }
15
16 public class aremain {
17
18     public static void main(String[] args) {
19         int l,b;
20         Scanner sc=new Scanner (System.in);
21         System.out.println("Enter the value of r");
22         r=sc.nextInt();
23         System.out.println("Enter the value of l");
24         l=sc.nextInt();
25         System.out.println("Enter the value of b");
26         b=sc.nextInt();
27         area a=new area();
28         a.area(l,b);
29         a.area(r);
30     }
31 }
```



The screenshot shows the Eclipse IDE with the file `dynamic.java` open. The Package Explorer on the left shows a project named `src` with a package `src` containing several Java files. The main editor displays the following code:

```
1 package IAN;
2 class a
3 {
4     void shaurya()
5     {
6         System.out.println("shaurya got placement");
7     }
8 }
9 class b extends a
10 {
11     void shaurya()
12     {
13         System.out.println("package of 10000");
14     }
15 }
16 public class dynamic
17 {
18     public static void main(String[] args)
19     {
20         a obj = new b();
21         obj.shaurya();
22     }
23 }
24 }
25 }
```



The screenshot shows the Eclipse IDE with the file `AccessAnimal.java` open. The Package Explorer on the left shows a project named `src` with a package `src` containing several Java files. The main editor displays the following code:

```
1 package IAN;
2
3 abstract class Animal {
4     // Field for the name of the animal
5     private String name;
6
7     // Constructor to get the name of the animal
8     public Animal(String name) {
9         this.name = name;
10    }
11
12    // Abstract method makeSound
13    public abstract void makeSound();
14
15    // Concrete method to get the name of the animal
16    public String getName() {
17        return name;
18    }
19
20    public void eat() {
21        System.out.println(getName()+" is eating food");
22    }
23 }
24
25 class Dog extends Animal {
26     public Dog(String name) {
27         super(name); // Call the constructor of the Animal class
28     }
29
30     // Implementing the abstract method for Dog
31     @Override
32     public void makeSound() {
33         System.out.println(getName() + " says: Bark");
34     }
35 }
36
37 class Cat extends Animal {
38     public Cat (String name) {
39         super(name); // Call the constructor of the Animal class
40     }
41 }
```



```
My eclipse-workspace - JavaCode\src\LAB\ExceptionMain.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JavaCode src (default package) 7 jul.java 8 ButtonDemo1.java 9 main.java 10 module-info.java 11 Chapter1 12 Ce.java 13 mainoverloading.java 14 i2.java 15 i3.java 16 Program1.java 17 i4pt.java 18 squareroot.java 19 sum.java 20 test.java 21 LAB 22 AccessAnimal.java 23 Animal.java 24 Counter.java 25 dynamic.java 26 EmployeeMain.java 27 ExceptionMain.java 28 FileHandling.java 29 InterfaceMain.java 30 Main.java 31 main703.java 32 mainoverloads.java 33 MyCalculatorMain.java 34 MyDialog.java 35 MyFrame.java 36 MyPanelDemo.java 37 i1.java 38 i2.java 39 package-info.java 40 salidrome.java 41 salidromeMain.java 42 sphere.java 43 ThreadMain.java
2 package IAH;
3 import java.io.BufferedReader;
4
5 public class ExceptionMain {
6
7     public static String isEven(int number) {
8         return number%2==0?"Even":"Odd";
9     }
10
11     public static void main(String[] args) throws Exception {
12         InputStreamReader isr = new InputStreamReader(System.in);
13         BufferedReader br = new BufferedReader(isr);
14         try {
15             System.out.println("Enter any number: ");
16             int x = Integer.parseInt(br.readLine());
17             System.out.println("!!!");
18             System.out.println(x+" is a "+isEven(x)+" number");
19         }
20         catch (Exception e) {
21             System.out.println(e);
22         }
23     }
24 }
25
26
27
```

```
My eclipse-workspace - JavaCode\src\LAB\FileHandling.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JavaCode src (default package) 7 jul.java 8 ButtonDemo1.java 9 main.java 10 module-info.java 11 Chapter1 12 Ce.java 13 mainoverloading.java 14 i2.java 15 i3.java 16 Program1.java 17 i4pt.java 18 squareroot.java 19 sum.java 20 test.java 21 LAB 22 AccessAnimal.java 23 Animal.java 24 Counter.java 25 dynamic.java 26 EmployeeMain.java 27 ExceptionMain.java 28 FileHandling.java 29 InterfaceMain.java 30 Main.java 31 main703.java 32 mainoverloads.java 33 MyCalculatorMain.java 34 MyDialog.java 35 MyFrame.java 36 MyPanelDemo.java 37 i1.java 38 i2.java 39 package-info.java 40 salidrome.java 41 salidromeMain.java 42 sphere.java 43 ThreadMain.java
1 package IAH;
2 import java.util.*;
3
4 public class FileHandling {
5
6     public static void main(String[] args) {
7         File dir = new File("IAH");
8         dir.mkdir();
9         File file = new File("LAB\\SS.txt");
10
11         try {
12             if (!file.createNewFile()) {
13                 System.out.println("File created at "+ file.getAbsolutePath());
14             } else {
15                 System.out.println("File already exists at "+ file.getAbsolutePath());
16             }
17         } catch (IOException e) {
18             System.out.println("Error found in creating the file");
19         }
20
21         try {
22             FileWriter writer = new FileWriter(file);
23             writer.write("Hi bla bla bla bla");
24             writer.close();
25         } catch (IOException e) {
26             System.out.println("Error found in writing the file");
27         }
28
29         try {
30             FileReader r = new FileReader(file);
31             int ch;
32             while ((ch = r.read()) != -1) {
33                 System.out.print((char) ch);
34             }
35             r.close();
36         } catch (IOException e) {
37             System.out.println("Error found in reading the file");
38         }
39     }
40 }
41
42
```

```
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer X
JavaCode
src
  default package
  7. java
  8. HelloWorldExample1.java
  9. main.java
  10. module-info.java
  11. Chapter1
    12. Cx.java
    13. mainoverloading.java
    14. i2.java
    15. i3.java
    16. Program1.java
    17. i4ptt.java
    18. i4squaremoot.java
    19. i4sum.java
    20. i4test.java
  21. i4nd
    22. i4AccessAnimal.java
    23. i4AccessMain.java
    24. i4BOOKS.java
    25. i4Counter.java
    26. i4dynamic.java
    27. i4EmployeeMain.java
    28. i4ExceptionMain.java
    29. i4FileHandling.java
    30. i4InterfaceMain.java
    31. i4Main.java
    32. i4main703.java
    33. i4mainoverloads.java
    34. i4MyCalculatorMain.java
    35. i4MyDialogue.java
    36. i4MyFrame.java
    37. i4MyPanelDemo.java
    38. i4i1.java
    39. i4i2.java
    40. i4package-info.java
    41. i4salldemo.java
    42. i4salldemoMain.java
    43. i4sphere.java
    44. i4ThreadMain.java
  45. dynamic.java
  46. EmployeeMain...
  47. ExceptionMa...
  48. FileHandling...
  49. Banki.java
  50. ArrayDemo.java
  51. Cx.java
  52. AlternateDe...
  53. Coverage
  54. ...

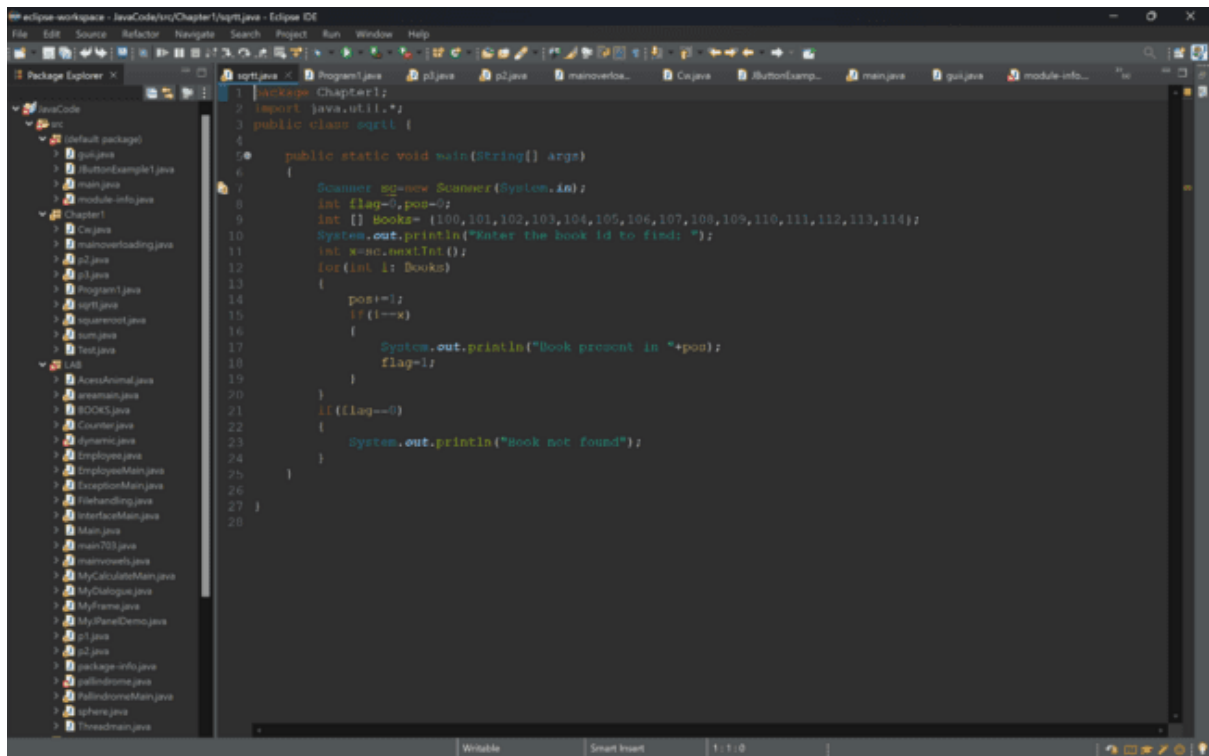
1 package IAH;
2 import java.util.*;
3
4
5
6 public class FileHandling {
7     public static void main(String[] args) {
8         File dir = new File("IAH");
9         dir.mkdir();
10        File file = new File("IAH\\SS.txt");
11
12        try {
13            if (file.createNewFile()) {
14                System.out.println("File created at" + file.getAbsolutePath());
15            } else {
16                System.out.println("File already exists at" + file.getAbsolutePath());
17            }
18        } catch (IOException e) {
19            System.out.println("Error found in creating the file");
20        }
21
22        try {
23            FileWriter writer = new FileWriter(file);
24            writer.write("Hi bla bla bla kkan");
25            writer.close();
26        } catch (IOException e) {
27            System.out.println("Error found in writing the file");
28        }
29
30        try {
31            FileReader r = new FileReader(file);
32            int ch;
33            while ((ch = r.read()) != -1) {
34                System.out.print((char) ch);
35            }
36            r.close();
37        } catch (IOException e) {
38            System.out.println("Error found in reading the file");
39        }
40    }
41}
42
```

```
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer X
JavaCode
src
  default package
  7. java
  8. HelloWorldExample1.java
  9. main.java
  10. module-info.java
  11. Chapter1
    12. Cx.java
    13. mainoverloading.java
    14. i2.java
    15. i3.java
    16. Program1.java
    17. i4ptt.java
    18. i4squaremoot.java
    19. i4sum.java
    20. i4test.java
  21. i4nd
    22. i4AccessAnimal.java
    23. i4AccessMain.java
    24. i4BOOKS.java
    25. i4Counter.java
    26. i4dynamic.java
    27. i4EmployeeMain.java
    28. i4ExceptionMain.java
    29. i4FileHandling.java
    30. i4InterfaceMain.java
    31. i4Main.java
    32. i4main703.java
    33. i4mainoverloads.java
    34. i4MyCalculatorMain.java
    35. i4MyDialogue.java
    36. i4MyFrame.java
    37. i4MyPanelDemo.java
    38. i4i1.java
    39. i4i2.java
    40. i4package-info.java
    41. i4salldemo.java
    42. i4salldemoMain.java
    43. i4sphere.java
    44. i4ThreadMain.java
  45. dynamic.java
  46. EmployeeMain...
  47. ExceptionMa...
  48. FileHandling...
  49. Banki.java
  50. ArrayDemo.java
  51. Cx.java
  52. AlternateDe...
  53. Coverage
  54. ...

1 package IAH;
2 import java.util.*;
3
4
5 class Emp
6 {
7     String name;
8     int eid,deptid;
9     void display(String name,int eid,int deptid)
10    {
11        this.name=name;
12        this.eid=eid;
13        this.deptid=deptid;
14        try
15        {
16            if(!((name.charAt(0))>=65 && name.charAt(0)<=90))
17            {
18                throw new ArithmeticException("The first letter is not Capital");
19            }
20        }
21        catch(Exception e)
22        {
23            System.out.println(e);
24        }
25        try
26        {
27            if(!((eid>=2001 && eid<=5001))
28            {
29                throw new Exception("The employee id id not in range of 2001 - 5001.");
30            }
31        }
32        catch(Exception e)
33        {
34            System.out.println(e);
35        }
36        try
37        {
38            if(!((deptid>=1 && deptid<=5))
39        }
40    }
41}
42
```

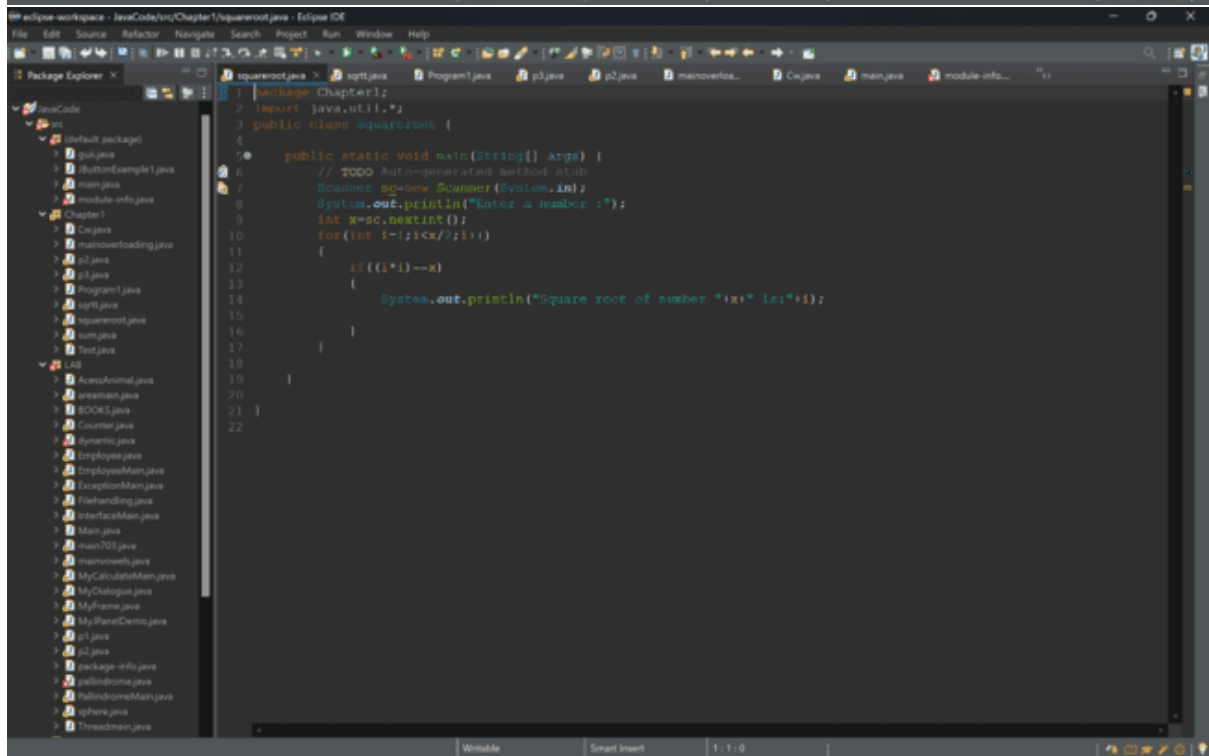
```
1 package IAN;
2
3
4 import java.util.*;
5 interface Demo
6 {
7     void volume();
8 }
9
10 class Cone implements Demo
11 {
12     double r,h,v;
13     Cone(double r,double h)
14     {
15         this.r=r;
16         this.h=h;
17     }
18
19     public void volume()
20     {
21         v=(float) (0.5*3.14*(r*r*h));
22         System.out.println("Volume of Cone:"+v);
23     }
24 }
25
26 class Hemisphere implements Demo
27 {
28     double r,v;
29     Hemisphere(double r)
30     {
31         this.r=r;
32     }
33
34     public void volume()
35     {
36         v=(double) (0.6666*3.14*(r*r*r));
37         System.out.println("Volume of Hemisphere:"+v);
38     }
39 }
40 class Cylinder implements Demo
```

```
29     Hemisphere(double r)
30     {
31         this.r=r;
32     }
33     public void volume()
34     {
35         v=(double) (0.6666*3.14*(r*r*r));
36         System.out.println("Volume of Hemisphere:"+v);
37     }
38 }
39
40 class Cylinder implements Demo
41 {
42     double r,h,v;
43     Cylinder(double r,double h)
44     {
45         this.r=r;
46         this.h=h;
47     }
48     public void volume()
49     {
50         v=(double) (3.14*(r*r*h));
51         System.out.println("Volume of Cylinder:"+v);
52     }
53 }
54
55 public class InterfaceMain {
56
57     public static void main(String[] args) {
58         Cone c=new Cone(3.0,1.0);
59         c.volume();
60         Hemisphere h=new Hemisphere(4.0);
61         h.volume();
62         Cylinder ci=new Cylinder(4.0,1.0);
63         ci.volume();
64     }
65 }
66
67 }
68
```



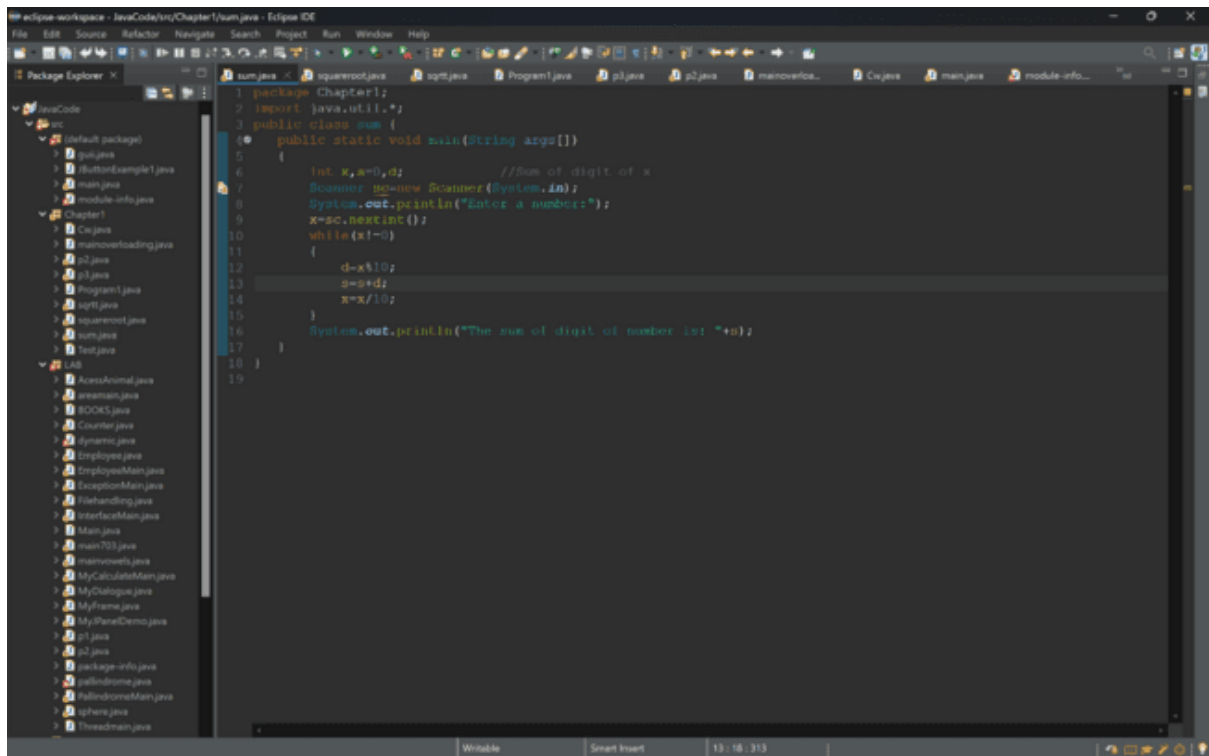
The screenshot shows the Eclipse IDE with the Package Explorer on the left displaying the project structure. The main editor window shows the code for `sqrrt.java` in the `Chapter1` package. The code implements a linear search algorithm to find a book ID in an array.

```
1 package Chapter1;
2 import java.util.*;
3 public class sqrrt {
4
5     public static void main(String[] args)
6     {
7         Scanner sc=new Scanner(System.in);
8         int flag=0,pos=-1;
9         int [] Books= {100,101,102,103,104,105,106,107,108,109,110,111,112,113,114};
10        System.out.println("Enter the book id to find: ");
11        int x=sc.nextInt();
12        for(int i: Books)
13        {
14            pos++;
15            if(i==x)
16            {
17                System.out.println("Book present in "+pos);
18                flag=1;
19            }
20        }
21        if(flag==0)
22        {
23            System.out.println("Book not found");
24        }
25    }
26 }
27 }
```

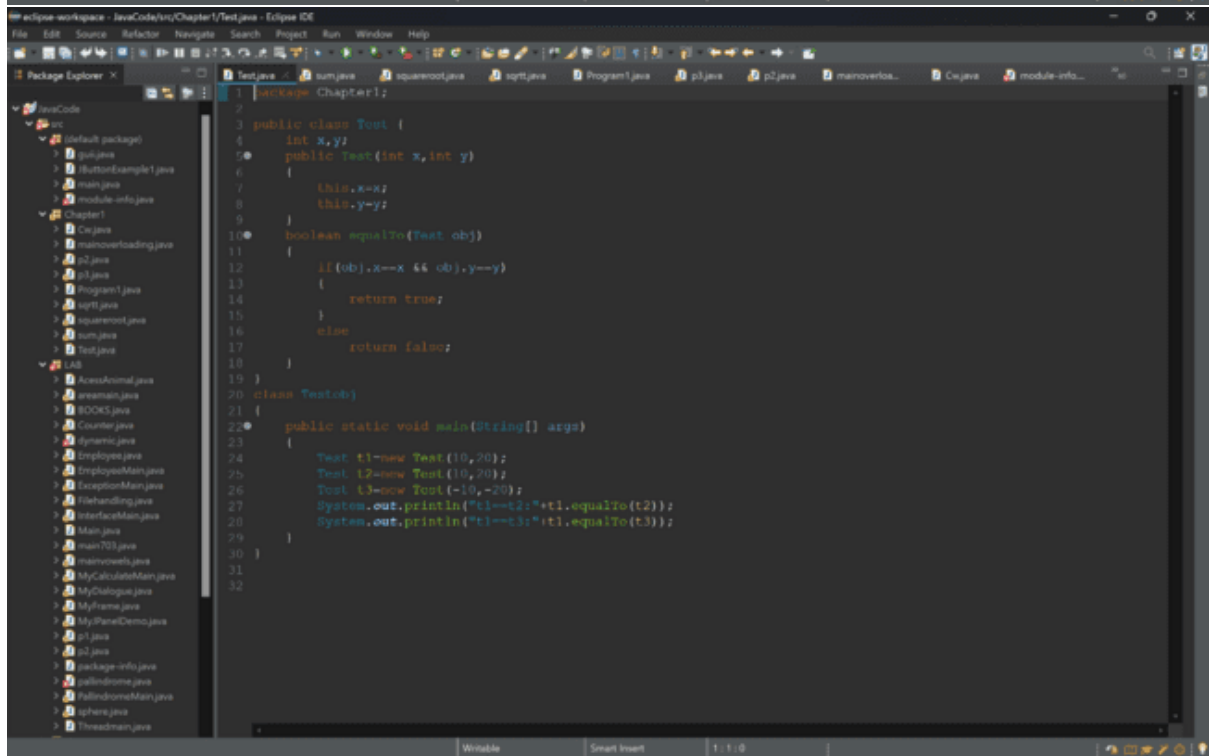


The screenshot shows the Eclipse IDE with the Package Explorer on the left displaying the project structure. The main editor window shows the code for `squareroot.java` in the `Chapter1` package. The code implements a program to calculate the square root of a number using a loop.

```
1 package Chapter1;
2 import java.util.*;
3 public class squareroot {
4
5     public static void main(String[] args) {
6         // TODO Auto-generated method stub
7         Scanner sc=new Scanner(System.in);
8         System.out.println("Enter a number :");
9         int n=sc.nextInt();
10        for(int i=1;i<=n/2;i++)
11        {
12            if(((i*i)==n))
13            {
14                System.out.println("Square root of number "+n+" is:"+i);
15            }
16        }
17    }
18 }
19 }
```



```
1 package Chapter1;
2 import java.util.*;
3 public class sum {
4     public static void main(String args[])
5     {
6         int x,a=0,d; //Sum of digit of x
7         Scanner sc=new Scanner(System.in);
8         System.out.println("Enter a number:");
9         x=sc.nextInt();
10        while(x!=0)
11        {
12            d=x%10;
13            s=s+d;
14            x=x/10;
15        }
16        System.out.println("The sum of digit of number is: "+s);
17    }
18 }
19
```



```
1 package Chapter1;
2
3 public class Test {
4     int x,y;
5     public Test(int x,int y)
6     {
7         this.x=x;
8         this.y=y;
9     }
10    boolean equalsTo(Test ob)
11    {
12        if((ob.x==x && ob.y==y))
13        {
14            return true;
15        }
16        else
17            return false;
18    }
19 }
20 class Testcd{
21 {
22     public static void main(String[] args)
23     {
24         Test t1=new Test(10,20);
25         Test t2=new Test(10,20);
26         Test t3=new Test(-10,-20);
27         System.out.println("t1==t2: "+t1.equalsTo(t2));
28         System.out.println("t1==t3: "+t1.equalsTo(t3));
29     }
30 }
31 }
32
```

```

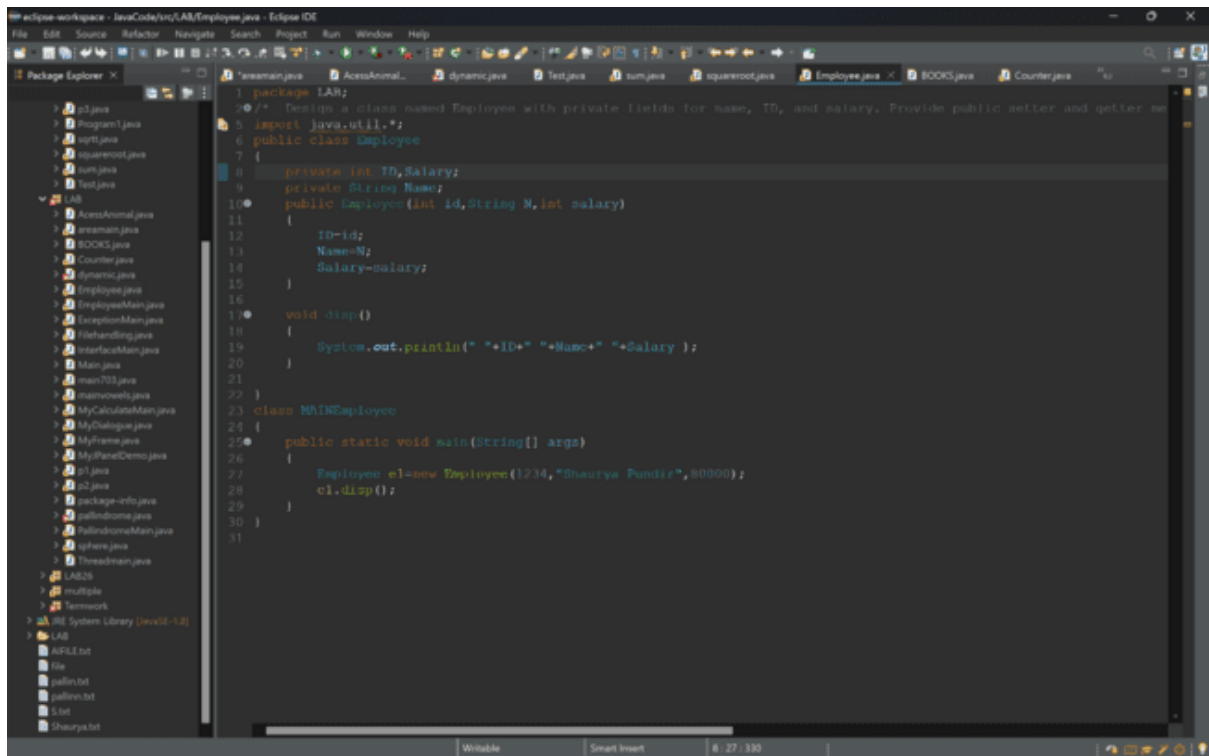
1 package IAN;
2 import java.util.*;
3
4 class area {
5     void area(int l,int b)
6     {
7         System.out.println("The area of rect is:"+l*b);
8     }
9     void area(int r)
10    {
11        System.out.println("The area of circle is:"+3.14*r*r);
12    }
13 }
14
15 public class areamain {
16
17     public static void main(String[] args) {
18         int x,l,b;
19         Scanner sc=new Scanner (System.in);
20         System.out.println("Enter the value of r");
21         r=sc.nextInt();
22         System.out.println("Enter the value of l");
23         l=sc.nextInt();
24         System.out.println("Enter the value of b");
25         b=sc.nextInt();
26         area a=new area();
27         a.area(l,b);
28         a.area(r);
29     }
30 }
31

```

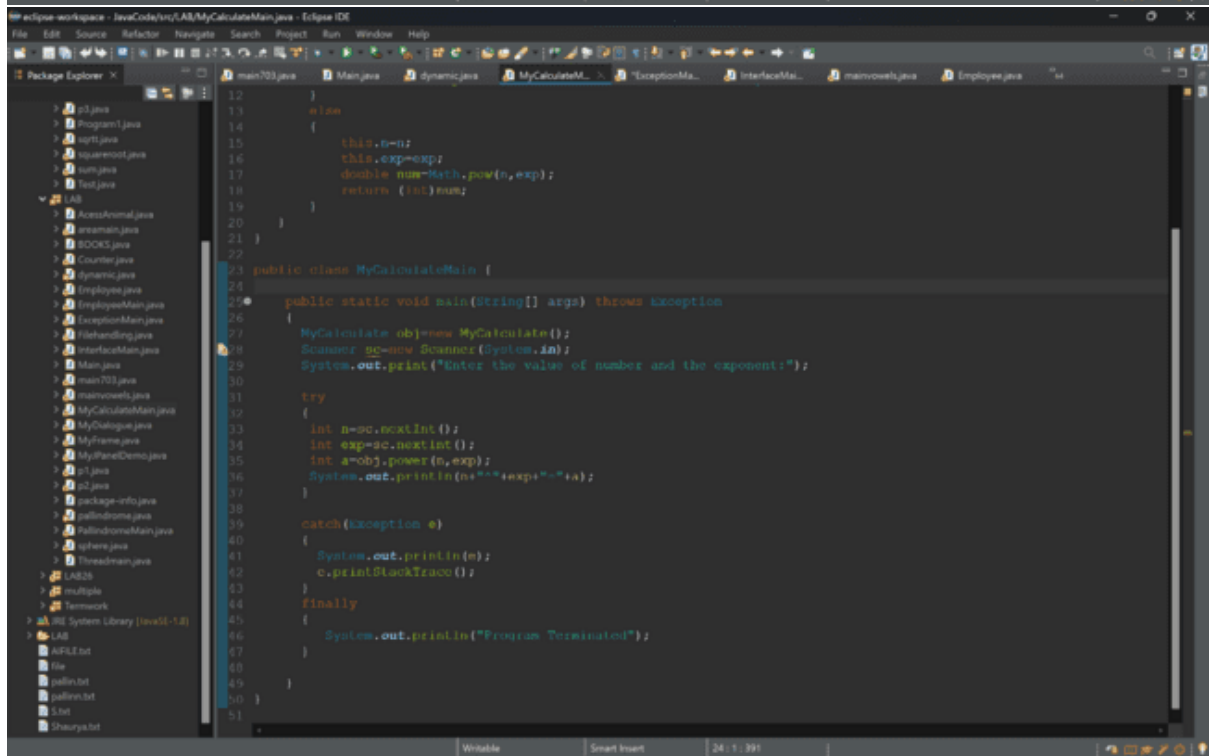
```

1 package IAN;
2
3 public class Counter
4 {
5     static int n=0;
6
7     public Counter()
8     {
9         n++;
10    }
11    void disp()
12    {
13        System.out.println("The number of times the instance created is :"+n);
14    }
15 }
16
17 class Countmain
18 {
19     public static void main(String args[])
20     {
21         Counter c1=new Counter();
22         Counter c2=new Counter();
23         Counter c3=new Counter();
24         c1.disp();
25     }
26 }
27
28

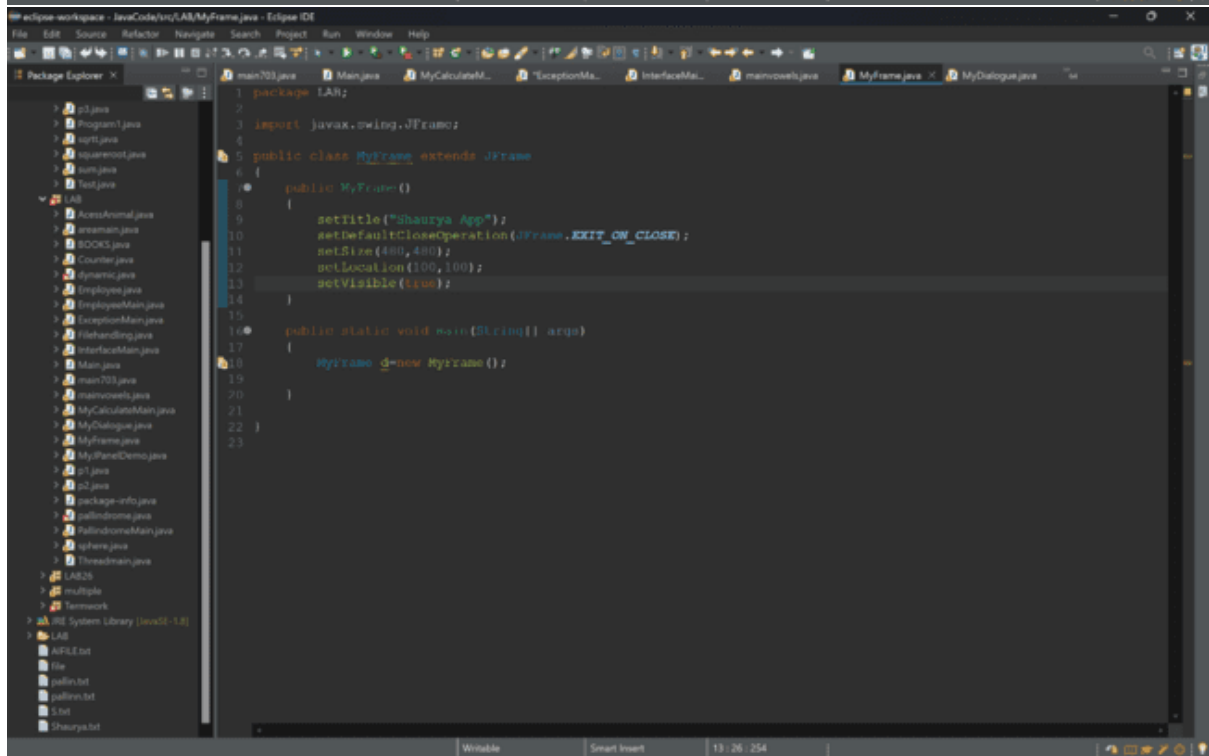
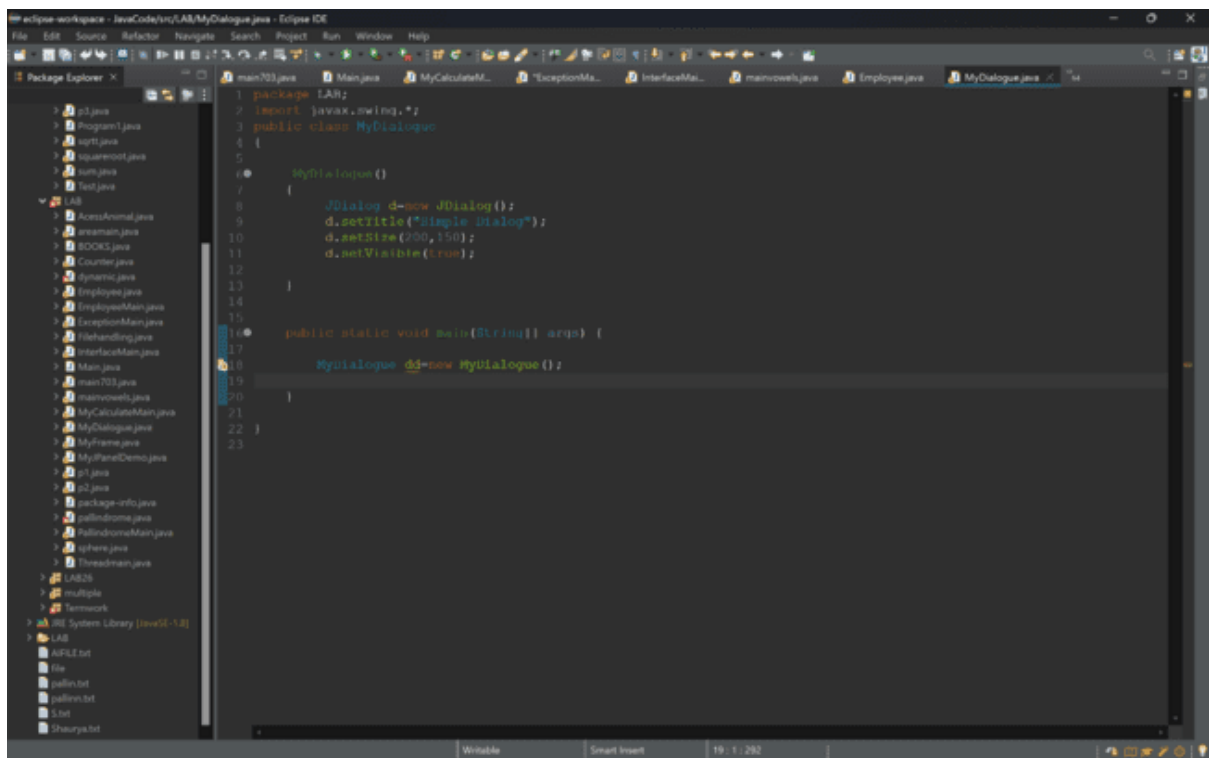
```

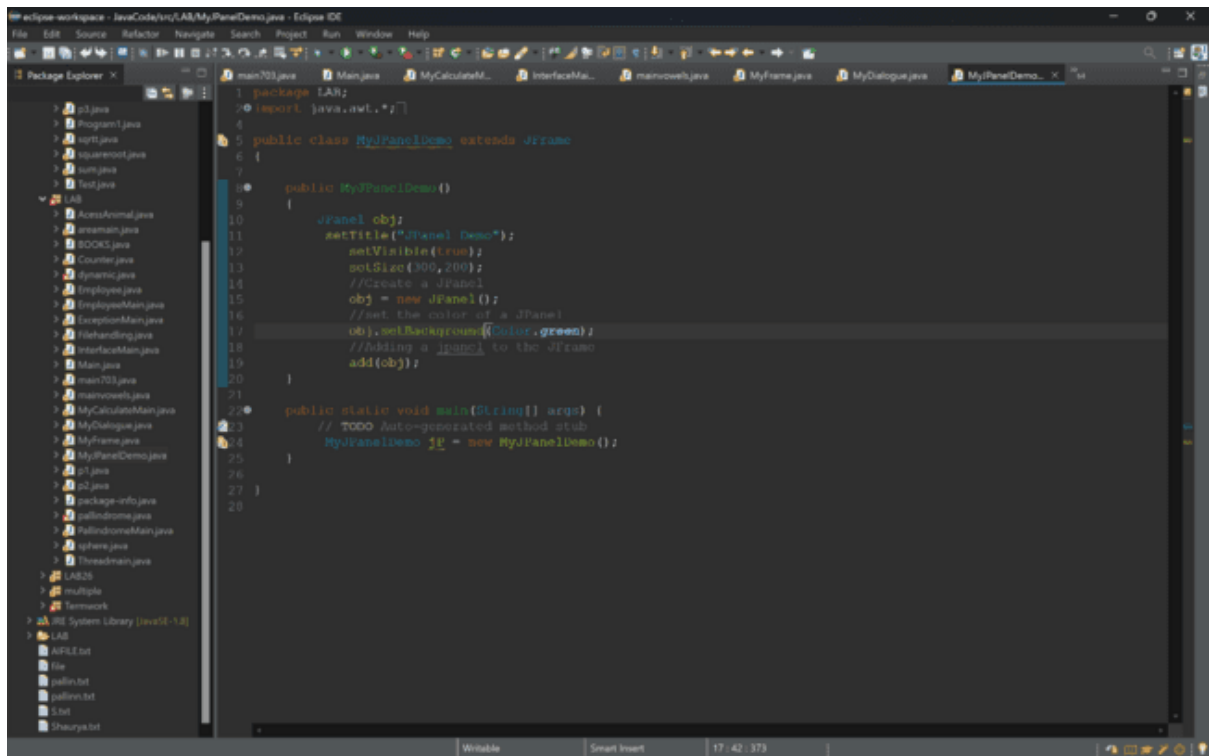



```
1 package LAB;
2
3 /**
4  * Designs a class named Employee with private fields for name, ID, and salary. Provide public setter and getter methods.
5  */
6 public class Employee
7 {
8     private int ID, Salary;
9     private String Name;
10    public Employee(int id, String N, int salary)
11    {
12        ID=id;
13        Name=N;
14        Salary=salary;
15    }
16
17    void disp()
18    {
19        System.out.println(" ID=" + ID + " Name=" + Name + " Salary = " + Salary);
20    }
21
22 }
23
24 class MAINEmployee
25 {
26     public static void main(String[] args)
27     {
28         Employee e1=new Employee(1234,"Shourya Pundir",80000);
29         e1.disp();
30     }
31 }
```



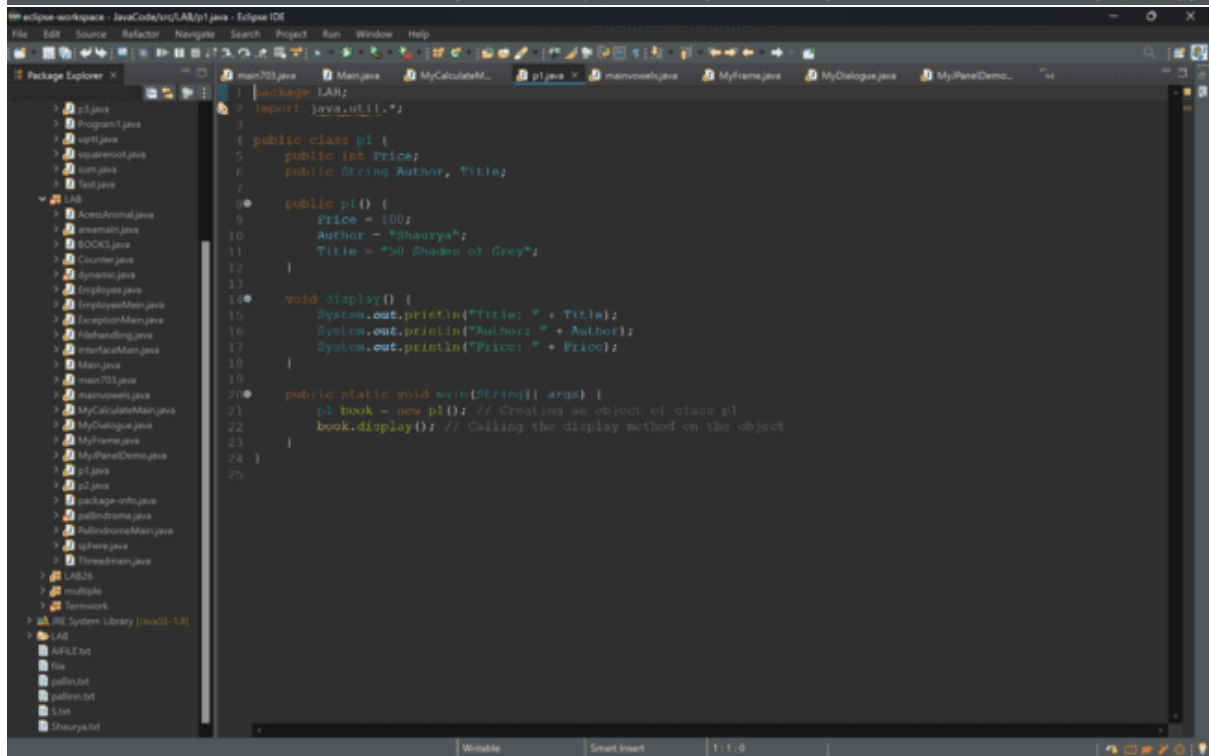
```
12 }
13
14 class
15 {
16     this.n=n;
17     this.exp=exp;
18     double num=Math.pow(n,exp);
19     return (int)num;
20 }
21
22
23 public class MyCalculateMain {
24
25     public static void main(String[] args) throws Exception
26     {
27         MyCalculate obj=new MyCalculate();
28         Scanner gc=new Scanner(System.in);
29         System.out.print("Enter the value of number and the exponent:");
30
31         try
32         {
33             int n=gc.nextInt();
34             int exp=gc.nextInt();
35             int a=obj.power(n,exp);
36             System.out.println(n+"^"+exp+"="+a);
37         }
38
39         catch(Exception e)
40         {
41             System.out.println(e);
42             e.printStackTrace();
43         }
44         finally
45         {
46             System.out.println("Program Terminated");
47         }
48     }
49 }
50
51 }
```



The screenshot shows the Eclipse IDE with the Package Explorer on the left and the Editor on the right. The Package Explorer shows a project named 'LAB' with a package 'LAB'. The Editor displays the code for 'MyJPanelDemo.java'.

```
1 package LAB;
2 import java.awt.*;
3
4
5 public class MyJPanelDemo extends JFrame
6 {
7
8     public MyJPanelDemo()
9     {
10         JPanel obj;
11         setTitle("JPanel Demo");
12         setVisible(true);
13         setSize(300,200);
14         //Create a JPanel
15         obj = new JPanel();
16         //set the color of a JPanel
17         obj.setBackground(Color.green);
18         //Adding a JPanel to the JFrame
19         add(obj);
20     }
21
22     public static void main(String[] args) {
23         // TODO Auto-generated method stub
24         MyJPanelDemo jf = new MyJPanelDemo();
25     }
26 }
27
28
```



The screenshot shows the Eclipse IDE with the Package Explorer on the left and the Editor on the right. The Package Explorer shows a project named 'LAB' with a package 'LAB'. The Editor displays the code for 'p1.java'.

```
1 package LAB;
2 import java.util.*;
3
4 public class p1 {
5     public int Price;
6     public String Author, Title;
7
8     public p1() {
9         Price = 100;
10        Author = "Shaurya";
11        Title = "50 Shades of Grey";
12    }
13
14    void display() {
15        System.out.println("Title: " + Title);
16        System.out.println("Author: " + Author);
17        System.out.println("Price: " + Price);
18    }
19
20    public static void main(String[] args) {
21        p1 book = new p1(); // Creating an object of class p1
22        book.display(); // Calling the display method on the object
23    }
24 }
25
```

```
1 package IAN;
2 import java.util.*;
3 public class p2
4 {
5     private int Price;
6     private String Author, Title;
7
8     public p2()
9     {
10         Author = "Yogesh";
11         Title = "XXX";
12         Price = 1000;
13     }
14     public p2(int x)
15     {
16         Author = "Shaurya";
17         Title = "50 Shades of Grey";
18     }
19
20     void display() {
21         System.out.println("Title: " + Title);
22         System.out.println("Author: " + Author);
23         System.out.println("Price: " + Price);
24     }
25
26     public static void main(String[] args) {
27         p2 book = new p2(); // Creating an object of class p1
28         p2 book2 = new p2(1);
29         book.display(); // Calling the display method on the object
30         book2.display();
31     }
32 }
33
34
35
```

```
1 package IAN;
2 import java.util.*;
3 class p
4 {
5     boolean pwork(String str)
6     {
7         String rev="";
8         for(int i=str.length()-1;i>=0;i--)
9         {
10             char ch=str.charAt(i);
11             rev+=ch;
12         }
13         return str.equals(rev);
14     }
15 }
16 public class palindrome {
17
18     public static void main(String[] args)
19     {
20         p p1=new p();
21         System.out.println("Enter the string :");
22         Scanner sc=new Scanner(System.in);
23         String x=sc.nextLine();
24         if(p1.pwork(x)==true)
25         {
26             System.out.println("The strings are palindrome");
27         }
28         else
29         {
30             System.out.println("The strings is not palindrome");
31         }
32     }
33 }
34
35
```

```

1 package LAB26;
2 import java.util.*;
3
4 class p {
5     String s;
6     p(String st) {
7         s = st;
8     }
9     String convert() {
10         StringBuilder result = new StringBuilder();
11         for (int i = 0; i < s.length(); i++) {
12             char ch = s.charAt(i);
13             if (ch >= 'A' && ch <= 'Z') {
14                 result.append(Character.toLowerCase(ch));
15             } else if (ch >= 'a' && ch <= 'z') {
16                 result.append(Character.toUpperCase(ch));
17             } else {
18                 result.append(ch);
19             }
20         }
21         return result.toString();
22     }
23 }
24
25 public class pluppermain {
26     static Scanner sc = new Scanner(System.in);
27     public static void main(String[] args) {
28         System.out.println("Enter the string:");
29         String s = sc.nextLine();
30         p p1 = new p(s);
31         String n = p1.convert();
32         System.out.println("The new string is: " + n);
33     }
34 }
35
```

```

1 package LAB26;
2
3 abstract class Shape {
4     int l, b, r;
5     float r2;
6     Shape(int a, int b) {
7         l=a;
8         this.b=b;
9     }
10    Shape(int c) {
11        this.c=c;
12    }
13    Shape(float r) {
14        this.r=r;
15    }
16    abstract void area();
17    // abstract void square_area();
18    // abstract void circle_area();
19 }
20
21 class Rectangle extends Shape {
22     Rectangle(int a1, int b1) {
23         super(a1, b1);
24     }
25     void area() {
26         System.out.println("The area of rectangle is: "+l*b);
27     }
28 }
29
30 class Square extends Shape {
31     Square(int a) {
32         super(a);
33     }
34 }
35
```

```
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer X
LAB
  AccessAnimal.java
  InvenMain.java
  BOOKS.java
  Counter.java
  Dynamic.java
  Employee.java
  EmployeeMain.java
  ExceptionMain.java
  FishHandling.java
  InterfaceMain.java
  Main.java
  main703.java
  maininsects.java
  MyCalculatorMain.java
  MyDialogue.java
  MyFrame.java
  MyPanelDemo.java
  r1.java
  r2.java
  package-info.java
  palindrome.java
  PalindromeMain.java
  sphere.java
  Threadmain.java
  MMain.java
  p3upperman.java
  package-info.java
  ShapeMain.java
  TemperatureMain.java
  volumeCalculator.java
multiple
Termwork
JRE System Library [JVM 9.0.4]
LAB
  APLE.txt
  file
  pullin.txt
  pullin1.txt
  E.txt
  Shourya.txt
package-info... p3upperman... sphere.java ShapeMain.java PalindromeM... "palindrome... Threadmain.java MMain.java "sa
34
35 }
36 class Square extends Shape
37 {
38     Square(int a)
39     {
40         super(a);
41     }
42     void area()
43     {
44         System.out.println("The are of square is :"+a*a);
45     }
46 }
47 class Circle extends Shape
48 {
49     Circle(float a)
50     {
51         super(a);
52     }
53     void area()
54     {
55         System.out.println("The are of square is :"+2*3.14*a);
56     }
57 }
58
59 public class ShapeMain {
60
61     public static void main(String[] args)
62     {
63         Rectangle r=new Rectangle(10,5);
64         Square s=new Square(10);
65         Circle c=new Circle((float) 7.0);
66         r.area();
67         s.area();
68         c.area();
69     }
70 }
71
72 }
73
Writeable Smart Insert 62 / 15 / 845
```

```
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer X
LAB
  AccessAnimal.java
  InvenMain.java
  BOOKS.java
  Counter.java
  Dynamic.java
  Employee.java
  EmployeeMain.java
  ExceptionMain.java
  FishHandling.java
  InterfaceMain.java
  Main.java
  main703.java
  maininsects.java
  MyCalculatorMain.java
  MyDialogue.java
  MyFrame.java
  MyPanelDemo.java
  r1.java
  r2.java
  package-info.java
  palindrome.java
  PalindromeMain.java
  sphere.java
  Threadmain.java
  MMain.java
  p3upperman.java
  package-info.java
  ShapeMain.java
  TemperatureMain.java
  volumeCalculator.java
multiple
Termwork
JRE System Library [JVM 9.0.4]
LAB
  APLE.txt
  file
  pullin.txt
  pullin1.txt
  E.txt
  Shourya.txt
package-info... p3upperman... sphere.java ShapeMain.java TemperatureM... PalindromeM... Threadmain.java MMain.java "sa
1 package LAB26;
2
3 import java.util.Scanner;
4
5 abstract class Temperature {
6     double temp;
7
8     void settemp(double temp1) {
9         temp = temp1;
10     }
11
12     abstract void changetemp();
13 }
14
15 class Faren_to_Celci extends Temperature {
16     Faren_to_Celci(double a) {
17         this.settemp(a);
18     }
19
20     void changetemp() {
21         temp = (5.0 / 9) * (temp - 32);
22         this.settemp(temp);
23         System.out.println("Fahrenheit to Celsius: " + temp);
24     }
25 }
26
27 class Celci to Faren extends Temperature {
28     Celci to Faren(double a) {
29         this.settemp(a);
30     }
31
32     void changetemp() {
33         temp = (5.0 / 9) * temp + 32;
34         this.settemp(temp);
35         System.out.println("Celsius to Fahrenheit: " + temp);
36     }
37 }
38
39 public class TemperatureMain {
40     public static void main(String[] args) {
41
Writeable Smart Insert 41 / 36 / 874
```

```

11 eclipse-workspace - JavaCode/tn/LAB26/TemperatureMain.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
LAB
  AccessAnimal.java
  Account.java
  Counter.java
  Employee.java
  EmployeeMain.java
  ExceptionMain.java
  FishHandling.java
  InterfaceMain.java
  Main.java
  main703.java
  main vowels.java
  MyCalculatorMain.java
  MyDialogues.java
  MyFrame.java
  MyPanelDemo.java
  1.java
  2.java
  package-info.java
  palindromeMain.java
  sphere.java
  ThreadMain.java
LAB26
  LABMain.java
  plupperman.java
  package-info.java
  ShapeMain.java
  temperatureMain.java
  volumeCalculator.java
multiple
Termwork
JRE System Library (jvarkit-1.8)
LAB
  APLE.txt
  file
  pullin.txt
  pullin.txt
  E.txt
  Shaurya.txt

21 temp = (5.0 / 9) * (temp - 32);
22 this.settemp(temp);
23 System.out.println("Fahrenheit to Celsius: " + temp);
24 }
25 }
26
27 class Celci_to_Faren extends Temperature {
28     Celci_to_Faren(double a) {
29         this.settemp(a);
30     }
31
32     void changetemp() {
33         temp = (5.0 / 9) * temp + 32;
34         this.settemp(temp);
35         System.out.println("Celsius to Fahrenheit: " + temp);
36     }
37 }
38
39 public class TemperatureMain {
40     public static void main(String[] args) {
41         Scanner sc = new Scanner(System.in);
42
43         // Convert Fahrenheit to Celsius
44         System.out.print("Enter temperature in Fahrenheit: ");
45         double fahrenheit = sc.nextDouble();
46         Faren_to_Celci farenToCelci = new Faren_to_Celci(fahrenheit);
47         farenToCelci.changetemp();
48
49         // Convert Celsius to Fahrenheit
50         System.out.print("Enter temperature in Celsius: ");
51         double celsius = sc.nextDouble();
52         Celci_to_Faren celciToFaren = new Celci_to_Faren(celsius);
53         celciToFaren.changetemp();
54
55         sc.close();
56     }
57 }
58
59
60

```

```

11 eclipse-workspace - JavaCode/tn/LAB26/VolumeCalculator.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
LAB
  AccessAnimal.java
  Account.java
  Counter.java
  Employee.java
  EmployeeMain.java
  ExceptionMain.java
  FishHandling.java
  InterfaceMain.java
  Main.java
  main703.java
  main vowels.java
  MyCalculatorMain.java
  MyDialogues.java
  MyFrame.java
  MyPanelDemo.java
  1.java
  2.java
  package-info.java
  palindromeMain.java
  sphere.java
  ThreadMain.java
LAB26
  LABMain.java
  plupperman.java
  package-info.java
  ShapeMain.java
  temperatureMain.java
  volumeCalculator.java
multiple
Termwork
JRE System Library (jvarkit-1.8)
LAB
  APLE.txt
  file
  pullin.txt
  pullin.txt
  E.txt
  Shaurya.txt

1 package LAB26;
2
3 import java.util.Scanner;
4
5 // Define the interface
6 interface Shapee {
7     void displayVolume();
8 }
9
10 // Class for Cone
11 class Cone implements Shapee {
12     private double radius;
13     private double height;
14
15     Cone(double radius, double height) {
16         this.radius = radius;
17         this.height = height;
18     }
19
20     //Override
21     public void displayVolume() {
22         double volume = (Math.PI * radius * radius * height) / 3;
23         System.out.println("Volume of the Cone: " + volume);
24     }
25 }
26
27 // Class for Hemisphere
28 class Hemisphere implements Shapee {
29     private double radius;
30
31     Hemisphere(double radius) {
32         this.radius = radius;
33     }
34
35     //Override
36     public void displayVolume() {
37         double volume = (2 * Math.PI * Math.pow(radius, 3)) / 3;
38         System.out.println("Volume of the Hemisphere: " + volume);
39     }
40 }

```

```

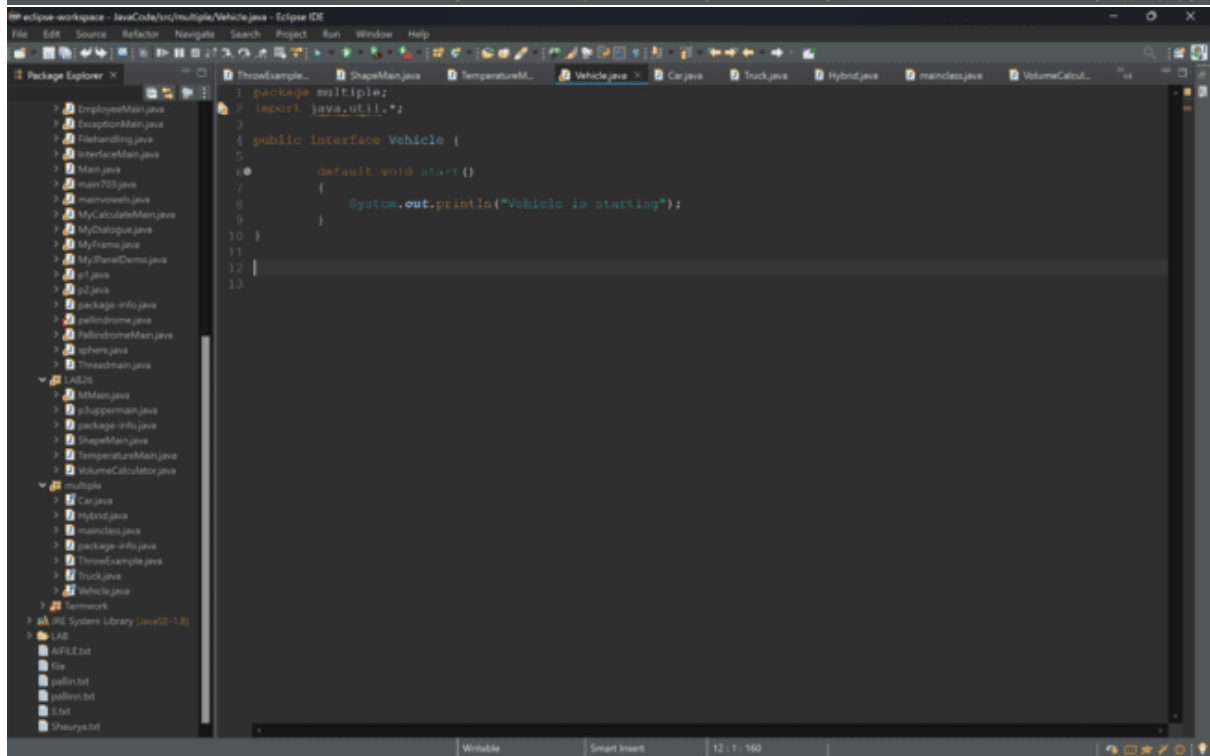
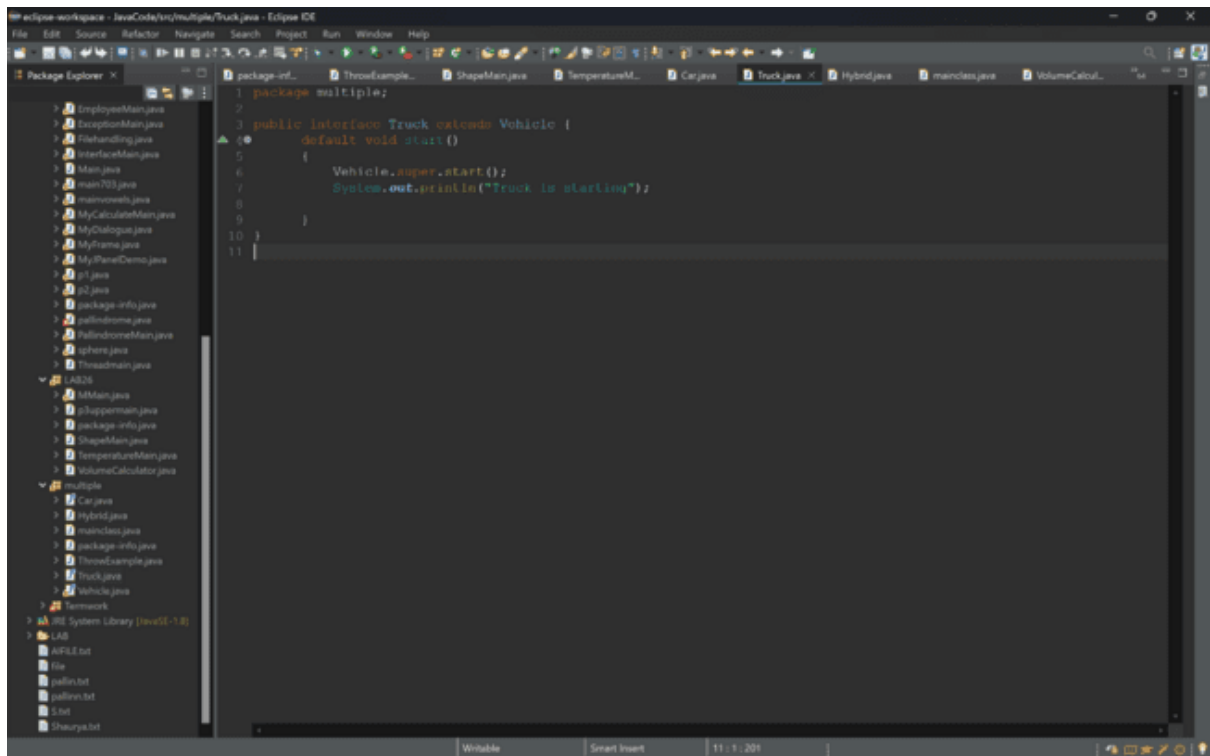
42 // Class for Cylinder
43 class Cylinder implements Shape {
44     private double radius;
45     private double height;
46
47     Cylinder(double radius, double height) {
48         this.radius = radius;
49         this.height = height;
50     }
51
52     @Override
53     public void displayVolume() {
54         double volume = Math.PI * radius * radius * height;
55         System.out.println("Volume of the cylinder: " + volume);
56     }
57 }
58
59 public class VolumeCalculator {
60     public static void main(String[] args) {
61         Scanner scanner = new Scanner(System.in);
62
63         // Get dimensions for Cone
64         System.out.print("Enter radius of the cone: ");
65         double coneRadius = scanner.nextDouble();
66         System.out.print("Enter height of the cone: ");
67         double coneHeight = scanner.nextDouble();
68         Cone cone = new Cone(coneRadius, coneHeight);
69
70         // Get dimensions for Hemisphere
71         System.out.print("Enter radius of the hemisphere: ");
72         double hemisphereRadius = scanner.nextDouble();
73         Hemisphere hemisphere = new Hemisphere(hemisphereRadius);
74
75         // Get dimensions for Cylinder
76         System.out.print("Enter radius of the cylinder: ");
77         double cylinderRadius = scanner.nextDouble();
78         System.out.print("Enter height of the cylinder: ");
79         double cylinderHeight = scanner.nextDouble();
80         Cylinder cylinder = new Cylinder(cylinderRadius, cylinderHeight);
81

```

```

1 package multiple;
2
3 public class ThrowExample {
4     static void checkAge(int age) throws Exception {
5         if (age < 18) {
6             // Throw an ArithmeticException if the age is less than 18
7             throw new Exception("Access denied - "
8                 + "You must be at least 18 years old.");
9         } else {
10             System.out.println("Access granted - You are old enough!");
11         }
12     }
13
14     public static void main(String[] args) {
15         try {
16             checkAge(10); // This will throw an ArithmeticException
17         } catch (Exception e) {
18             System.out.println("Exception caught in main: " + e);
19         }
20     }
21 }
22
23

```

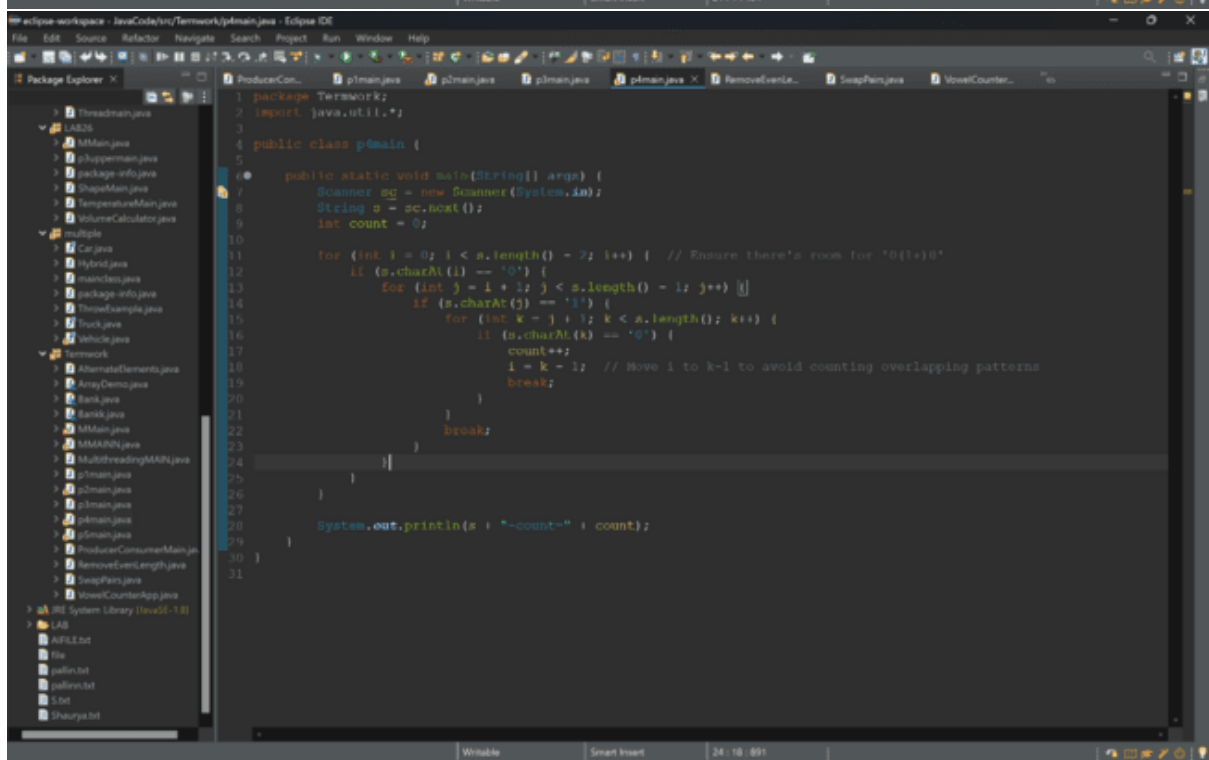


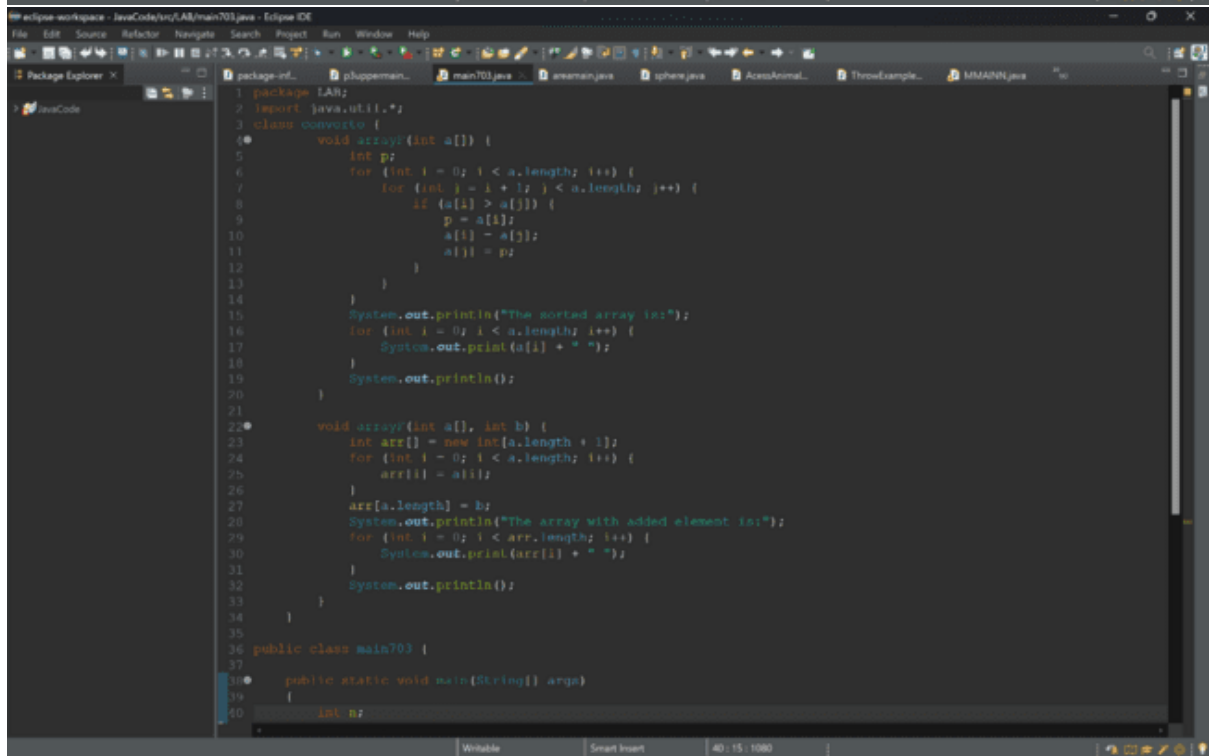
ThreadExample... | ProducerCon... | j5man.java | Vehicle.java | Car.java | Truck.java | Hybrid.java | mairclass.java | VolumeCalcul...

```
40 Thread.sleep(100); // Simulate time taken to produce a value
41 } catch (InterruptedException e) {
42     e.printStackTrace();
43 }
44 }
45 }
46 }
47
48 class Consumer extends Thread {
49     private SharedBuffer buffer;
50
51     public Consumer(SharedBuffer buffer) {
52         this.buffer = buffer;
53     }
54
55     @Override
56     public void run() {
57         for (int i = 1; i <= 10; i++) {
58             try {
59                 buffer.consume();
60                 Thread.sleep(150); // Simulate time taken to consume a value
61             } catch (InterruptedException e) {
62                 e.printStackTrace();
63             }
64         }
65     }
66 }
67
68 public class ProducerConsumerMain {
69     public static void main(String[] args) {
70         SharedBuffer buffer = new SharedBuffer();
71         Producer producer = new Producer(buffer);
72         Consumer consumer = new Consumer(buffer);
73
74         producer.start();
75         consumer.start();
76
77         try {
78             producer.join();
79             consumer.join();
80         }
```

ThreadExample... | ProducerCon... | j5man.java | Vehicle.java | Car.java | Truck.java | Hybrid.java | mairclass.java | VolumeCalcul...

```
47
48 class Consumer extends Thread {
49     private SharedBuffer buffer;
50
51     public Consumer(SharedBuffer buffer) {
52         this.buffer = buffer;
53     }
54
55     @Override
56     public void run() {
57         for (int i = 1; i <= 10; i++) {
58             try {
59                 buffer.consume();
60                 Thread.sleep(150); // Simulate time taken to consume a value
61             } catch (InterruptedException e) {
62                 e.printStackTrace();
63             }
64         }
65     }
66 }
67
68 public class ProducerConsumerMain {
69     public static void main(String[] args) {
70         SharedBuffer buffer = new SharedBuffer();
71         Producer producer = new Producer(buffer);
72         Consumer consumer = new Consumer(buffer);
73
74         producer.start();
75         consumer.start();
76
77         try {
78             producer.join();
79             consumer.join();
80         } catch (InterruptedException e) {
81             e.printStackTrace();
82         }
83     }
84 }
85
86 }
```





```
My eclipse-workspace - JavaCode\src\LAB\main703.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JavaCode
package-info package-info... phupperman... main703.java... areaman.java... sphere.java... AccessAnimal... ThrowExample... MMANN.java...
20 }
21
22 void arrayF(int a[], int b) {
23     int arr[] = new int[a.length + 1];
24     for (int i = 0; i < a.length; i++) {
25         arr[i] = a[i];
26     }
27     arr[a.length] = b;
28     System.out.println("The array with added element is:");
29     for (int i = 0; i < arr.length; i++) {
30         System.out.print(arr[i] + " ");
31     }
32     System.out.println();
33 }
34 }
35
36 public class main703 {
37
38     public static void main(String[] args)
39     {
40         int n;
41         Scanner sc = new Scanner(System.in);
42         System.out.println("Enter the value of n");
43         n = sc.nextInt();
44         int arr[] = new int[n];
45         System.out.println("Enter the Value in array");
46         for (int i = 0; i < n; i++)
47         {
48             arr[i] = sc.nextInt();
49         }
50         System.out.println("Enter the value of b");
51         int b = sc.nextInt();
52         convert c1 = new convert();
53         c1.arrayF(arr);
54         c1.arrayF(arr, b);
55     }
56 }
57
58 }
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```

```
1 package Termmork;
2
3
4 import java.util.ArrayList;
5
6
7 class Bank {
8     private static int nextAccountNumber = 1001;
9     private String name;
10    private String address;
11    private int accountNumber;
12    private double balance;
13
14    // Constructor to initialize the bank account
15    public Bank(String name, String address, double initialBalance) {
16        this.name = name;
17        this.address = address;
18        this.accountNumber = nextAccountNumber++;
19        this.balance = initialBalance;
20    }
21
22    // Method to display account information
23    public void display() {
24        System.out.println("Account Number: " + accountNumber);
25        System.out.println("Name: " + name);
26        System.out.println("Address: " + address);
27        System.out.println("Balance: " + balance);
28    }
29
30    // Method to deposit amount
31    public void deposit(double amount) {
32        if (amount > 0) {
33            balance += amount;
34            System.out.println("Deposited: " + amount);
35        } else {
36            System.out.println("Invalid deposit amount");
37        }
38    }
39
40    // Method to withdraw amount
41    public void withdraw(double amount) {
```

```
40    // Method to withdraw amount
41    public void withdraw(double amount) {
42        if (amount > 0 && amount <= balance) {
43            balance -= amount;
44            System.out.println("Withdrawn: " + amount);
45        } else {
46            System.out.println("Invalid or insufficient amount for withdrawal");
47        }
48    }
49
50    // Method to change address
51    public void changeAddress(String newAddress) {
52        address = newAddress;
53        System.out.println("Address updated to: " + newAddress);
54    }
55
56    // Getters for account number and name (for identifying accounts)
57    public int getAccountNumber() {
58        return accountNumber;
59    }
60
61    public String getName() {
62        return name;
63    }
64
65    public static void main(String[] args) {
66        Scanner scanner = new Scanner(System.in);
67        ArrayList<Bank> bankAccounts = new ArrayList<>();
68
69        // Enter the number of depositors
70        System.out.print("Enter the number of depositors: ");
71        int numDepositors = scanner.nextInt();
72        scanner.nextLine(); // Consume newline
73
74        // Enter information for each depositor
75        for (int i = 0; i < numDepositors; i++) {
76            System.out.print("Enter name of depositor: ");
77            String name = scanner.nextLine();
78            System.out.print("Enter address of depositor: ");
79            String address = scanner.nextLine();
```

```

73
74 // Enter information for each depositor
75 for (int i = 0; i < numDepositors; i++) {
76     System.out.print("Enter name of depositor: ");
77     String name = scanner.nextLine();
78     System.out.print("Enter address of depositor: ");
79     String Address = scanner.nextLine();
80     System.out.print("Enter initial balance: ");
81     double balance = scanner.nextDouble();
82     scanner.nextLine(); // Consume newline
83
84     Bankk account = new Bankk(name, Address, balance);
85     bankAccounts.add(account);
86     System.out.println("Account created with account number: " + account.getAccountNumber());
87 }
88
89 // Perform operations
90 boolean continueOperations = true;
91 while (continueOperations) {
92     System.out.println("\nChoose an operation:");
93     System.out.println("1. Display information of a depositor");
94     System.out.println("2. Deposit amount");
95     System.out.println("3. Withdraw amount");
96     System.out.println("4. Change address");
97     System.out.println("5. Exit");
98     int choice = scanner.nextInt();
99     scanner.nextLine(); // Consume newline
100
101     switch (choice) {
102         case 1:
103             System.out.print("Enter account number: ");
104             int accNumberToDisplay = scanner.nextInt();
105             scanner.nextLine(); // Consume newline
106             for (Bankk account : bankAccounts) {
107                 if (account.getAccountNumber() == accNumberToDisplay) {
108                     account.displayInfo();
109                 }
110             }
111             break;
112     }
113 }

```

```

1 package IAN;
2 import java.util.*;
3
4
5 class Emp
6 {
7     String name;
8     int eid,deptid;
9     void display(String name,int eid,int deptid)
10    {
11        this.name=name;
12        this.eid=eid;
13        this.deptid=deptid;
14        try
15        {
16            if(!name.charAt(0)>=65 && name.charAt(0)<=90))
17            {
18                throw new ArithmeticException("The first letter is not Capital");
19            }
20        }
21        catch(Exception e)
22        {
23            System.out.println(e);
24        }
25        try
26        {
27            if(!eid>=2001 && eid<=5001))
28            {
29                throw new Exception("The employee id is not in range of 2001 - 5001.");
30            }
31        }
32        catch(Exception e)
33        {
34            System.out.println(e);
35        }
36        try
37        {
38            if(!deptid>=1 && deptid<=5))
39            {
40

```

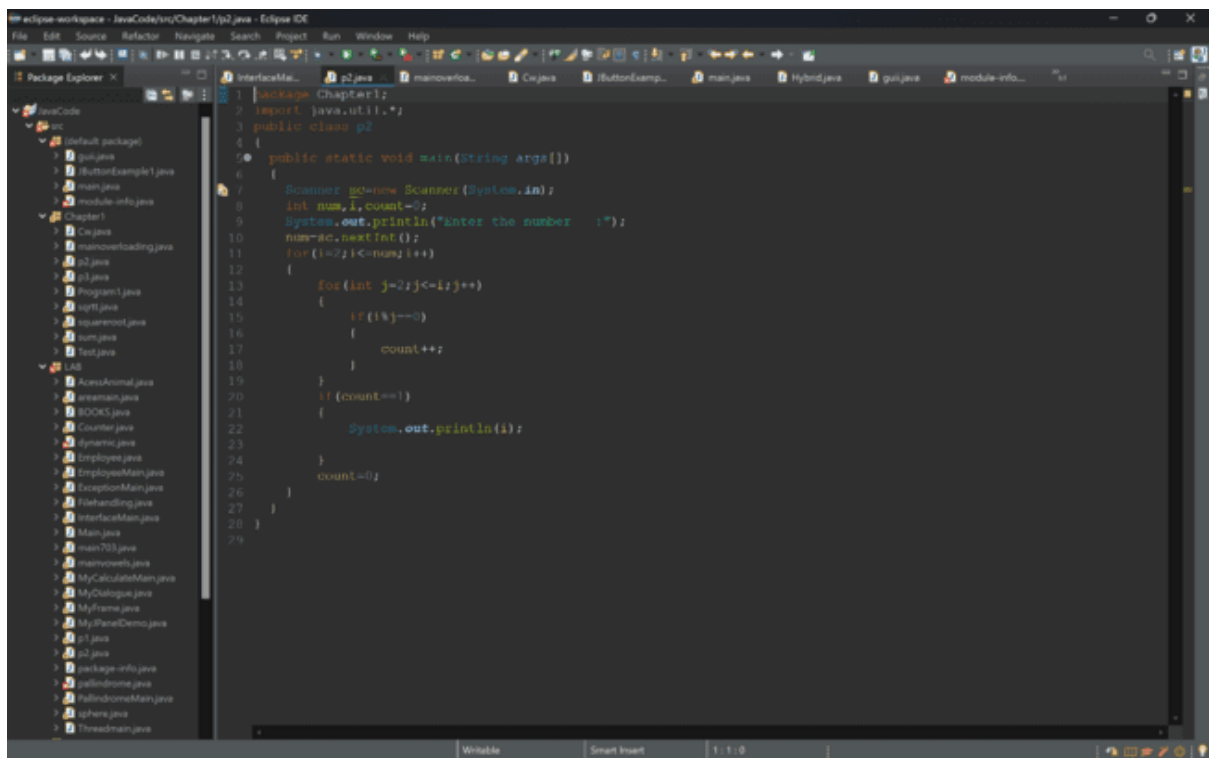
```
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JavaCode src (default package) EmployeeMain.java
30 throw new Exception("The employee id is not in range of 2001 - 5001.");
31
32 catch(Exception e)
33 {
34     System.out.println(e);
35 }
36 try
37 {
38     if(!deptId>=1 && deptId<=5)
39     {
40         throw new Exception("The dept id is not in the range of 1 - 5.");
41     }
42     catch(Exception e)
43     {
44         System.out.println(e);
45     }
46     System.out.println("Name :"+name+"dept id: "+deptId+"dept ID: "+deptId);
47 }
48 }
49 public class EmployeeMain {
50     public static void main(String[] args)
51     {
52         Scanner sc=new Scanner(System.in);
53         System.out.println("Enter the Employee name,Employee id and Department ID:");
54         String name=sc.next();
55         int eid=sc.nextInt();
56         int deptId=sc.nextInt();
57         Emp e=new Emp();
58         e.display(name,eid,deptId);
59     }
60 }
61
62 package Termwork;
63 import java.util.ArrayList;
64
65 public class AlternateElements {
66     public static List<Integer> alternate(List<Integer> list1, List<Integer> list2) {
67         List<Integer> result = new ArrayList<>();
68         int size1 = list1.size();
69         int size2 = list2.size();
70         int maxSize = Math.max(size1, size2);
71         for (int i = 0; i < maxSize; i++) {
72             if (i < size1)
73                 result.add(list1.get(i));
74             if (i < size2)
75                 result.add(list2.get(i));
76         }
77         return result;
78     }
79     public static void main(String[] args) {
80         List<Integer> list1 = new ArrayList<>();
81         list1.add(1);
82         list1.add(2);
83         list1.add(3);
84         list1.add(4);
85         list1.add(5);
86
87         List<Integer> list2 = new ArrayList<>();
88         list2.add(6);
89         list2.add(7);
90         list2.add(8);
91         list2.add(9);
92         list2.add(10);
93
94         List<Integer> result = alternate(list1, list2);
95         System.out.println(result);
96     }
97 }
```

```
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JavaCode src (default package) AlternateElements.java
1 package Termwork;
2 import java.util.ArrayList;
3
4 public class AlternateElements {
5     public static List<Integer> alternate(List<Integer> list1, List<Integer> list2) {
6         List<Integer> result = new ArrayList<>();
7         int size1 = list1.size();
8         int size2 = list2.size();
9         int maxSize = Math.max(size1, size2);
10         for (int i = 0; i < maxSize; i++) {
11             if (i < size1)
12                 result.add(list1.get(i));
13             if (i < size2)
14                 result.add(list2.get(i));
15         }
16         return result;
17     }
18     public static void main(String[] args) {
19         List<Integer> list1 = new ArrayList<>();
20         list1.add(1);
21         list1.add(2);
22         list1.add(3);
23         list1.add(4);
24         list1.add(5);
25
26         List<Integer> list2 = new ArrayList<>();
27         list2.add(6);
28         list2.add(7);
29         list2.add(8);
30         list2.add(9);
31         list2.add(10);
32
33         List<Integer> result = alternate(list1, list2);
34         System.out.println(result);
35     }
36 }
```

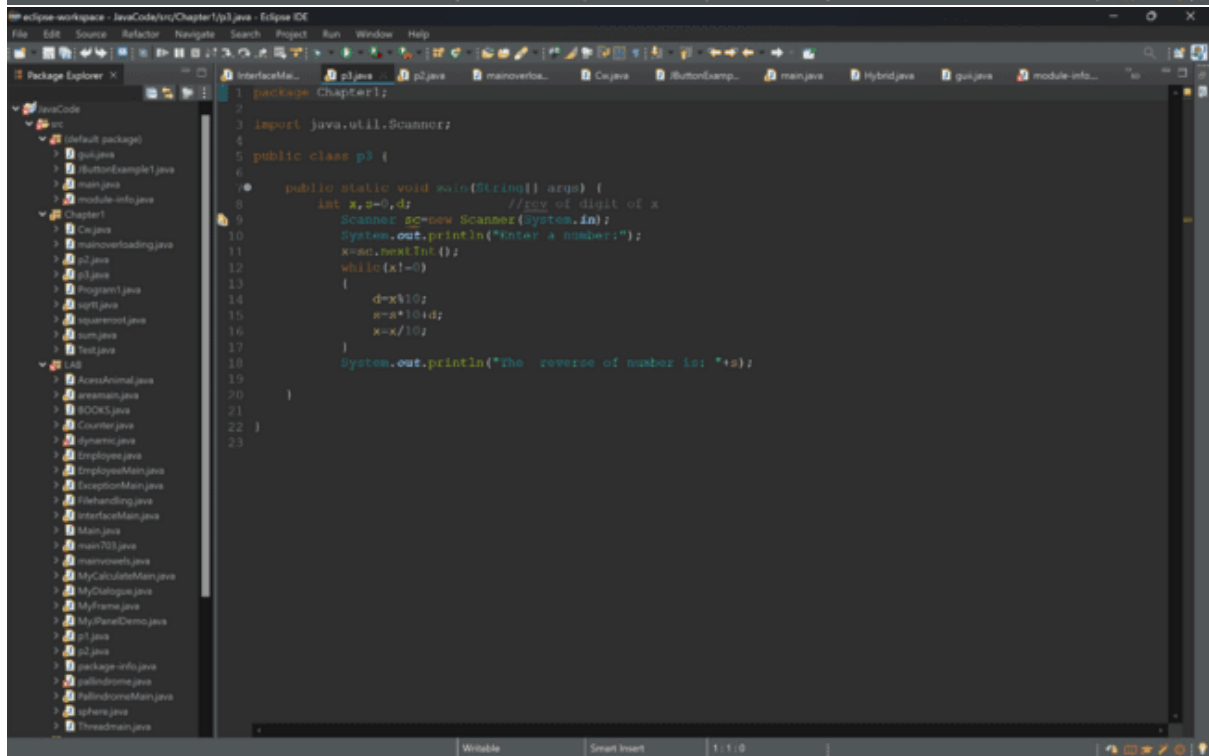


```
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JavaCode src (default package) 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
ArrayDemo.java
1 package TerminiWork;
2
3 import java.util.Arrays;
4
5 class ArrayDemo {
6
7     // Overloaded arrayFunc method to find pairs with a specific sum
8     void arrayFunc(int[] array, int target) {
9         System.out.println("Pairs of elements whose sum is " + target + " are:");
10        for (int i = 0; i < array.length; i++) {
11            for (int j = i + 1; j < array.length; j++) {
12                if (array[i] + array[j] == target) {
13                    System.out.println(array[i] + " + " + array[j] + " = " + target);
14                }
15            }
16        }
17    }
18
19    void arrayFunc(int[] A, int p, int[] B, int q) {
20        int[] merged = new int[p + q];
21        int i = 0, j = 0, k = 0;
22
23        // Merge the two arrays
24        while (i < p && j < q) {
25            if (A[i] < B[j]) {
26                merged[k++] = A[i++];
27            } else {
28                merged[k++] = B[j++];
29            }
30        }
31
32        // Copy remaining elements from A
33        while (i < p) {
34            merged[k++] = A[i++];
35        }
36
37        // Copy remaining elements from B
38        while (j < q) {
39            merged[k++] = B[j++];
40        }
41    }
42}
```

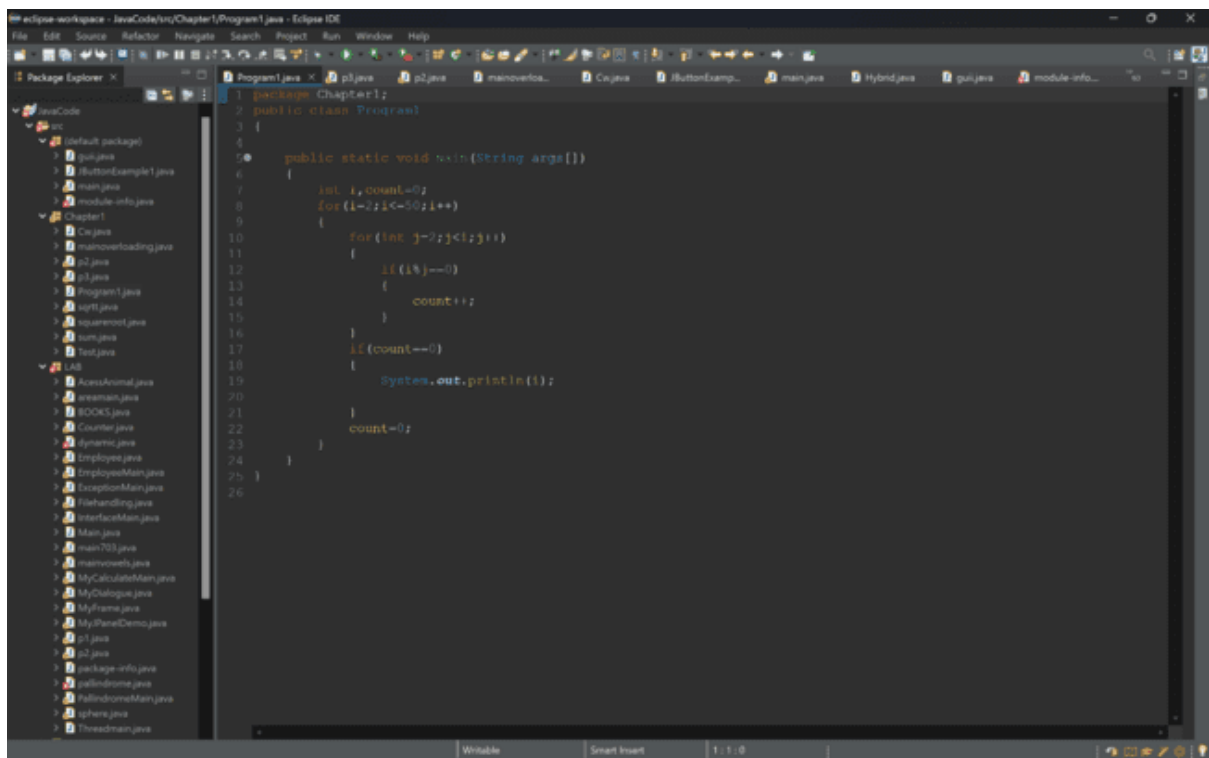
```
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JavaCode src (default package) 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
JButtonExample1.java
1 import javax.swing.*;
2
3 public class JButtonExample1 {
4
5     public static void main(String[] args) {
6         // Create a new JFrame
7         JFrame frame = new JFrame("JButton Example");
8         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
9         frame.setSize(400, 400);
10        frame.setLayout(new FlowLayout());
11
12        // JButton with no text or icon
13        JButton button1 = new JButton();
14        button1.setText("Submit");
15        button1.setToolTipText("This is Submit Button");
16        frame.add(button1);
17
18        // JButton with text
19        JButton button2 = new JButton("Reset");
20        button2.setToolTipText("This is a Reset Button");
21        frame.add(button2);
22
23        // Configure JButton using various methods
24        button1.setVerticalTextPosition(SwingConstants.BOTTOM);
25        button1.setHorizontalTextPosition(SwingConstants.CENTER);
26        button2.setVerticalTextPosition(SwingConstants.BOTTOM);
27        button2.setHorizontalTextPosition(SwingConstants.CENTER);
28
29        // button1.setMargin(new Insets(10, 10, 10, 10));
30        // button1.setRounded(true);
31        // button1.setCustomAreaFilled(true);
32        // button1.setBorderPainted(true);
33
34        // Make the frame visible
35        frame.setVisible(true);
36    }
37}
```



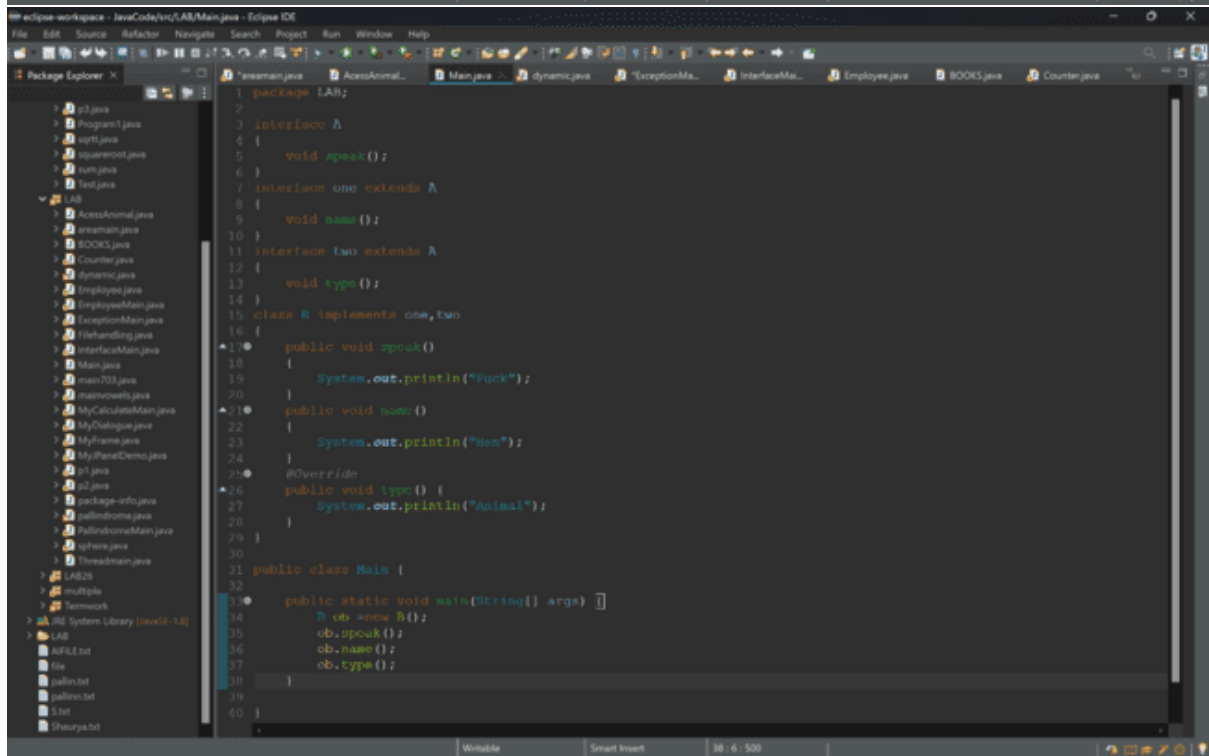
```
1 package Chapter1;
2 import java.util.*;
3 public class p2 {
4 {
5     public static void main(String args[])
6     {
7         Scanner sc=new Scanner(System.in);
8         int num,i,count=0;
9         System.out.println("Enter the number :");
10        num=sc.nextInt();
11        for(i=2;i<=num;i++)
12        {
13            for(int j=2;j<=i;j++)
14            {
15                if(i%j==0)
16                {
17                    count++;
18                }
19            }
20            if(count==1)
21            {
22                System.out.println(i);
23            }
24            count=0;
25        }
26    }
27 }
28 }
29 }
```



```
1 package Chapter1;
2
3 import java.util.Scanner;
4
5 public class p3 {
6 {
7     public static void main(String[] args) {
8         int x,s=0,d; //rev of digit of x
9         Scanner sc=new Scanner(System.in);
10        System.out.println("Enter a number:");
11        x=sc.nextInt();
12        while(x!=0)
13        {
14            d=x%10;
15            s=s*10+d;
16            x=x/10;
17        }
18        System.out.println("The reverse of number is: "+s);
19    }
20 }
21 }
22 }
23 }
```



```
1 package Chapter1;
2 public class Program
3 {
4
5     public static void main(String args[])
6     {
7         int i, count=0;
8         for(i=2; i<=50; i++)
9         {
10             for(int j=2; j<=i; j++)
11             {
12                 if(i%j==0)
13                 {
14                     count++;
15                 }
16             }
17             if(count==0)
18             {
19                 System.out.println(i);
20             }
21             count=0;
22         }
23     }
24 }
25
26
```



```
1 package LAB;
2
3 interface IAnimal
4 {
5     void speak();
6 }
7 interface IAnimalTwo extends IAnimal
8 {
9     void name();
10 }
11 interface IAnimalThree extends IAnimalTwo
12 {
13     void type();
14 }
15 class B implements IAnimal, IAnimalTwo, IAnimalThree
16 {
17     public void speak()
18     {
19         System.out.println("Buck");
20     }
21     public void name()
22     {
23         System.out.println("Bee");
24     }
25     @Override
26     public void type() {
27         System.out.println("Animal");
28     }
29 }
30
31 public class Main {
32
33     public static void main(String[] args) {
34         B ob = new B();
35         ob.speak();
36         ob.name();
37         ob.type();
38     }
39 }
40

```

```

1 package LAB;
2
3 interface A {
4     void speak();
5 }
6
7 interface one extends A {
8     void name();
9 }
10
11 interface two extends A {
12     void type();
13 }
14
15 class B implements one, two {
16     public void speak() {
17         System.out.println("Puck");
18     }
19     public void name() {
20         System.out.println("Men");
21     }
22     @Override
23     public void type() {
24         System.out.println("Animal");
25     }
26 }
27
28 public class Main {
29     public static void main(String[] args) {
30         B ob = new B();
31         ob.speak();
32         ob.name();
33         ob.type();
34     }
35 }

```

```

1 package LAB;
2 import java.util.*;
3
4 class vowels {
5     int countvowels(String str) {
6         int count = 0;
7         for (int i = 0; i < str.length(); i++) { // Corrected loop condition
8             char ch = str.charAt(i);
9             if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
10                 count++;
11             }
12         }
13         return count;
14     }
15 }
16
17 public class mainvowels {
18     public static void main(String args[]) {
19         vowels vl = new vowels();
20         Scanner sc = new Scanner(System.in);
21         System.out.println("Enter the String:");
22         String x = sc.nextLine();
23         int co = vl.countvowels(x);
24         System.out.println("No. of vowels in " + x + " is: " + co); // Added space after "is"
25     }
26 }

```

```
MyEclipse-workspace - JavaCode/tn/LAB/MyCalculateMain.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer X
1 package LAB;
2 import java.util.*;
3
4 class MyCalculate
5 {
6     int n,exp;
7     int power(int n,int exp) throws Exception
8     {
9         if(n<0 || exp<0)
10            {
11                throw new Exception("The number n & n should be non negative");
12            }
13         else
14            {
15                this.n=n;
16                this.exp=exp;
17                double num=Math.pow(n,exp);
18                return (int)num;
19            }
20     }
21 }
22
23 public class MyCalculateMain {
24
25     public static void main(String[] args) throws Exception
26     {
27         MyCalculate obj=new MyCalculate();
28         Scanner s=new Scanner(System.in);
29         System.out.print("Enter the value of number and the exponent:");
30
31         try
32         {
33             int n=s.nextInt();
34             int exp=s.nextInt();
35             int a=obj.power(n,exp);
36             System.out.println(n+"^"+exp+"="+a);
37         }
38         catch (Exception e)
39         {
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```
try {
    FileWriter writer = new FileWriter(file);
    writer.write("Hello madan how are you");
    writer.close();
} catch (IOException e) {
    System.out.println("Error found in writing the file");
    e.printStackTrace();
}

try {
    FileReader fileReader = new FileReader(file);
    BufferedReader bufferedReader = new BufferedReader(fileReader);
    String line;
    int palindromeCount = 0;
    while ((line = bufferedReader.readLine()) != null) {
        String[] words = line.split("\\s");
        for (String word : words) {
            if (isPalindrome(word)) {
                palindromeCount++;
            }
        }
    }
    bufferedReader.close();
    System.out.println("Number of palindrome words: " + palindromeCount);
} catch (IOException e) {
    System.out.println("Error found in reading the file");
    e.printStackTrace();
}
}

//write a java prog to create a interface and make classes cone, cylinder and hemisphere that implements the interfa
//write a java file handlini prog to count and display the no of palindrome present in the text file*/
```

```
package LAB;

class ThreadCount extends Thread {
    private int threadno;
    public ThreadCount(int threadno) {
        this.threadno=threadno;
    }
    public void run() {
        try {
            countThread();
        } catch (InterruptedException e) {
            e.printStackTrace();
        }
    }
    public void countThread() throws InterruptedException {
        for(int i=0; i<5; i++) {
            Thread.sleep(500);
            System.out.println("Thread No. "+threadno+" iteration "+i);
        }
    }
}

//write a java file handlini prog to count and display the no of palindrome present in the text file*/
public class Threadmain {
    public static void main(String[] args) throws InterruptedException {
        long startTime=System.currentTimeMillis();
        for(int i=0; i<10; i++) {
            ThreadCount t1=new ThreadCount(i);
            t1.start();
        }
        Thread.sleep(3000);
    }
}
```

```
MyEclipse-workspace - JavaCode\src\LAB\Threadmain.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer X
  java
  > Program1.java
  > sptt.java
  > sptttest.java
  > sum.java
  > test.java
  > LAB
    > AccessAnimal.java
    > AccessMain.java
    > BOOKS.java
    > Counter.java
    > dynamic.java
    > Employee.java
    > EmployeeMain.java
    > ExceptionMain.java
    > FishHandling.java
    > InterfaceMain.java
    > Main.java
    > main703.java
    > main703test.java
    > MyCalculatorMain.java
    > MyDialog.java
    > MyFrame.java
    > MyPanelDemo.java
    > s1.java
    > s2.java
    > package-info.java
    > palindrome.java
    > PalindromeMain.java
    > sphere.java
    > Threadmain.java
  > LAB26
  > multiple
  > Termwork
  > JRE System Library [jvarkit-1.8]
  > LAB
    > APLE.txt
    > file
    > pullin.txt
    > pullin.txt
    > s.txt
    > Shourya.txt

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public void run()
{
    try {
        countThread();
    } catch (InterruptedException e) {
        e.printStackTrace();
    }
}

public void countThread() throws InterruptedException
{
    for(int i=0;i<5;i++)
    {
        Thread.sleep(500);
        System.out.println("Thread No. "+Thread.currentThread().getName()+" iteration "+i);
    }
}

//write a java file handling program to count and display the no of palindrome present in the text file?
public class Threadmain {
    public static void main(String[] args) throws InterruptedException
    {
        long startTime=System.currentTimeMillis();
        for(int i=0;i<10;i++)
        {
            ThreadCount t1=new ThreadCount();
            t1.start();
        }
        Thread.sleep(3000);
        System.out.println("Main thread is exiting.....");
        long endTime=System.currentTimeMillis();
        System.out.println("Time taken: "+(endTime-startTime)/1000+" seconds");
    }
}
```

```
MyEclipse-workspace - JavaCode\src\LAB26\VolumeCalculator.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer X
  java
  > Program1.java
  > sptt.java
  > sptttest.java
  > sum.java
  > test.java
  > LAB
    > AccessAnimal.java
    > AccessMain.java
    > BOOKS.java
    > Counter.java
    > dynamic.java
    > Employee.java
    > EmployeeMain.java
    > ExceptionMain.java
    > FishHandling.java
    > InterfaceMain.java
    > Main.java
    > main703.java
    > main703test.java
    > MyCalculatorMain.java
    > MyDialog.java
    > MyFrame.java
    > MyPanelDemo.java
    > s1.java
    > s2.java
    > package-info.java
    > palindrome.java
    > PalindromeMain.java
    > sphere.java
    > Threadmain.java
  > LAB26
  > multiple
  > Termwork
  > JRE System Library [jvarkit-1.8]
  > LAB
    > APLE.txt
    > file
    > pullin.txt
    > pullin.txt
    > s.txt
    > Shourya.txt

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@Override
public void displayVolume() {
    double volume = Math.PI * radius * radius * height;
    System.out.println("Volume of the Cylinder: " + volume);
}

public class VolumeCalculator {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Get dimensions for Cone
        System.out.print("Enter radius of the cone: ");
        double coneradius = scanner.nextDouble();
        System.out.print("Enter height of the cone: ");
        double coneheight = scanner.nextDouble();
        Cone cone = new Cone(coneradius, coneheight);

        // Get dimensions for hemisphere
        System.out.print("Enter radius of the hemisphere: ");
        double hemisphereRadius = scanner.nextDouble();
        Hemisphere hemisphere = new Hemisphere(hemisphereRadius);

        // Get dimensions for Cylinder
        System.out.print("Enter radius of the cylinder: ");
        double cylinderRadius = scanner.nextDouble();
        System.out.print("Enter height of the cylinder: ");
        double cylinderheight = scanner.nextDouble();
        Cylinder cylinder = new Cylinder(cylinderRadius, cylinderheight);

        // Display volumes
        System.out.println();
        cone.displayVolume();
        hemisphere.displayVolume();
        cylinder.displayVolume();

        scanner.close();
    }
}
```


My eclipse-workspace - JavaCode\src\multiple\Car.java - Eclipse IDE

File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer

- EmployeeMain.java
- ExceptionMain.java
- InterfaceMain.java
- Main.java
- Main703.java
- MyCalculatorMain.java
- MyDialogMain.java
- MyFrame.java
- MyPanelDemo.java
- Test.java
- Test2.java
- package-info.java
- Rollingstone.java
- RollingstoneMain.java
- Sphere.java
- ThreadMain.java
- LAB25
 - MyMain.java
 - ShopperMain.java
 - package-info.java
 - ShapeMain.java
 - TemperatureMain.java
 - VolumeCalculator.java
- multiple
 - Car.java
 - Hybrid.java
 - Mainclass.java
 - package-info.java
 - ThreadExample.java
 - Truck.java
 - Vehicle.java
- Termwork
- IDE System Library [JDK-1.8]
 - LAB
 - APLLE.txt
 - file
 - pullin.txt
 - pullinv.txt
 - Test
 - Thaurya.txt

```
1 package multiple;
2
3 public interface Car extends Vehicle
4 {
5     default void start()
6     {
7         Vehicle.super.start();
8         System.out.println("Car is starting");
9     }
10 }
11
```

Window | Smart Insert | 3 / 37 | 57

My eclipse-workspace - JavaCode\src\multiple\Hybrid.java - Eclipse IDE

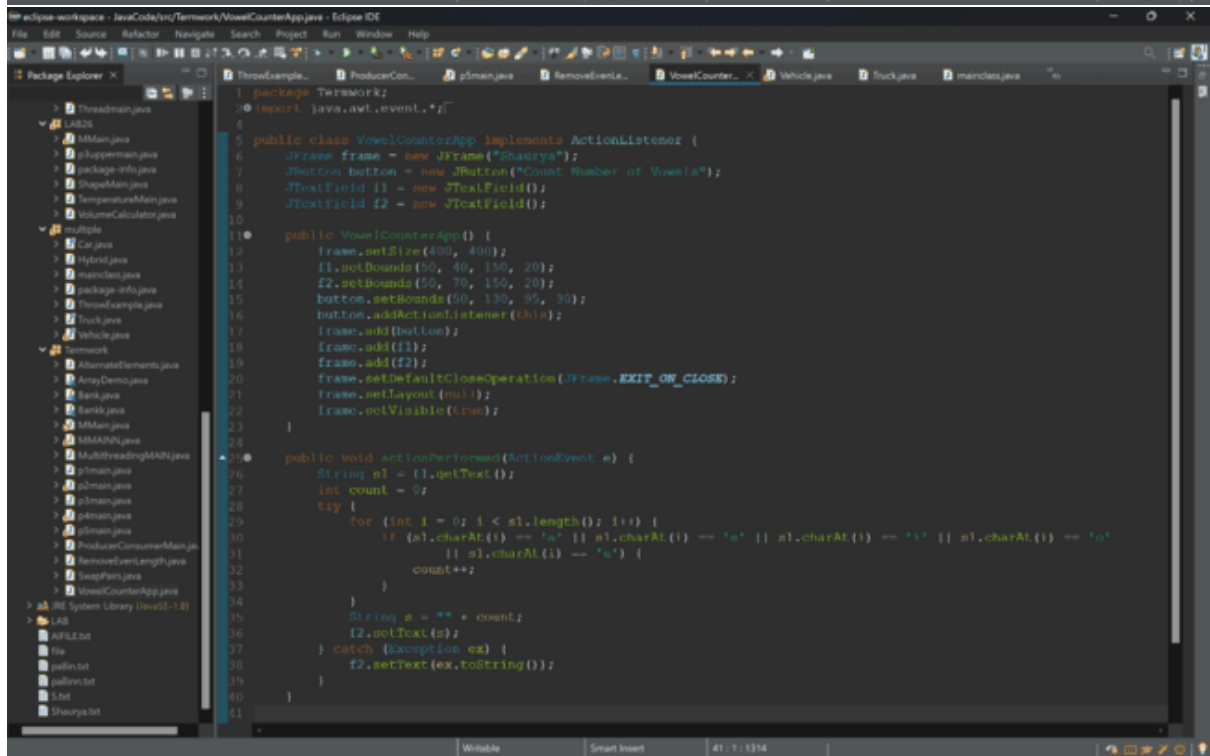
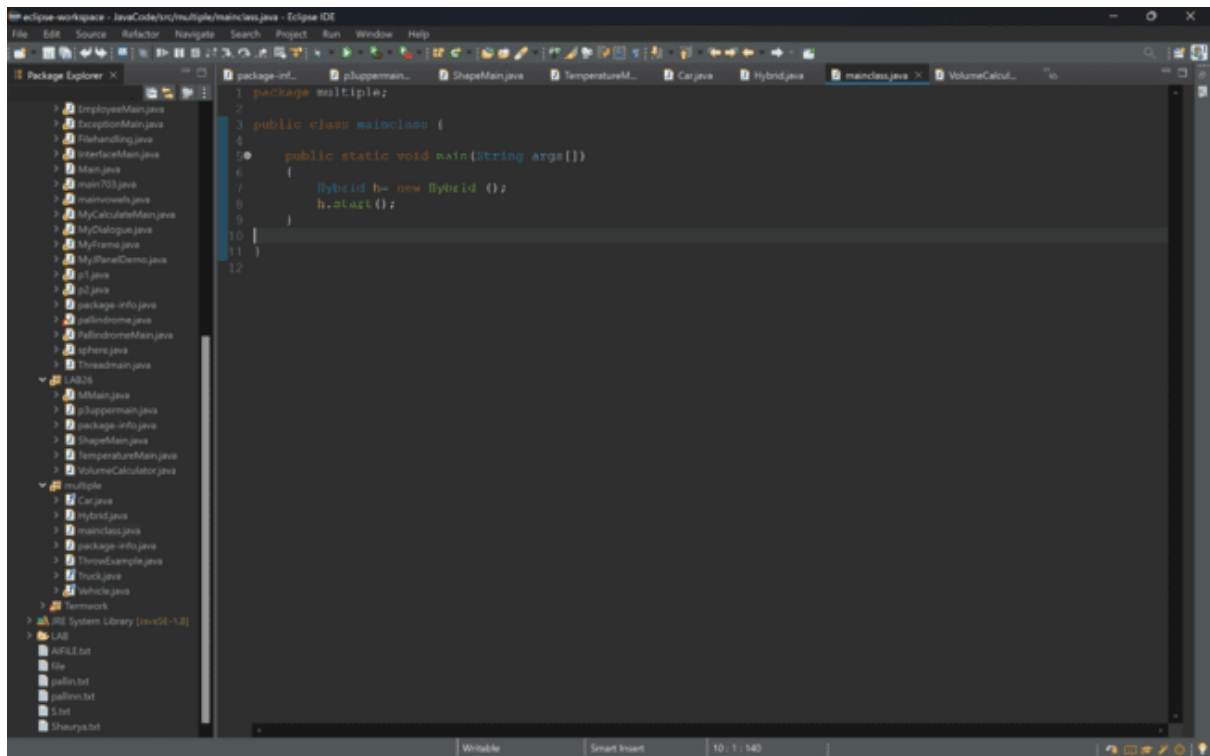
File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer

- EmployeeMain.java
- ExceptionMain.java
- InterfaceMain.java
- Main.java
- Main703.java
- MyCalculatorMain.java
- MyDialogMain.java
- MyFrame.java
- MyPanelDemo.java
- Test.java
- Test2.java
- package-info.java
- Rollingstone.java
- RollingstoneMain.java
- Sphere.java
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- LAB25
 - MyMain.java
 - ShopperMain.java
 - package-info.java
 - ShapeMain.java
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 - Car.java
 - Hybrid.java
 - Mainclass.java
 - package-info.java
 - ThreadExample.java
 - Truck.java
 - Vehicle.java
- Termwork
- IDE System Library [JDK-1.8]
 - LAB
 - APLLE.txt
 - file
 - pullin.txt
 - pullinv.txt
 - Test
 - Thaurya.txt

```
1 package multiple;
2
3 public class Hybrid implements Car, Truck {
4
5     public void start()
6     {
7         System.out.println("Hybrid class ");
8         Car.super.start();
9         Truck.super.start();
10    }
11
12 }
13
```

Window | Smart Insert | 11 / 1 : 190



```
1 package Termwork;
2
3 import java.util.ArrayList;
4
5 public class SwapPairs {
6
7     public static void swapPairs(ArrayList<String> list) {
8         for (int i = 0; i < list.size() - 1; i += 2) {
9             // Swap the elements at indices i and i+1
10             String temp = list.get(i);
11             list.set(i, list.get(i + 1));
12             list.set(i + 1, temp);
13         }
14     }
15
16     public static void main(String[] args) {
17         ArrayList<String> list1 = new ArrayList<>();
18         list1.add("four");
19         list1.add("score");
20         list1.add("and");
21         list1.add("seven");
22         list1.add("years");
23         list1.add("ago");
24
25         System.out.println("Original list: " + list1);
26         swapPairs(list1);
27         System.out.println("List after swapping pairs: " + list1);
28
29         ArrayList<String> list2 = new ArrayList<>();
30         list2.add("a");
31         list2.add("b");
32         list2.add("c");
33         list2.add("d");
34         list2.add("e");
35         list2.add("f");
36         list2.add("g");
37         list2.add("h");
38
39         System.out.println("Original list: " + list2);
40         swapPairs(list2);
41         System.out.println("List after swapping pairs: " + list2);
42     }
43 }
```

```
8 Here, 7 is the saddle point because it is the minimum element in its row and maximum element in its column.
9 */
10 public class p1main {
11     public static void main(String[] args) {
12         int[][] matrix = {
13             {1, 2, 3},
14             {4, 5, 6},
15             {7, 8, 9}
16         };
17
18         findSaddlePoint(matrix);
19     }
20
21     public static void findSaddlePoint(int[][] matrix) {
22         int rows = matrix.length;
23         int cols = matrix[0].length;
24
25         for (int i = 0; i < rows; i++) {
26             int minRowValue = matrix[i][0];
27             int minColIndex = 0;
28
29             // Find the minimum element in the current row
30             for (int j = 1; j < cols; j++) {
31                 if (matrix[i][j] < minRowValue) {
32                     minRowValue = matrix[i][j];
33                     minColIndex = j;
34                 }
35             }
36
37             boolean isSaddlePoint = true;
38
39             // Check if the found minimum element is the maximum in its column
40             for (int k = 0; k < rows; k++) {
41                 if (matrix[k][minColIndex] > minRowValue) {
42                     isSaddlePoint = false;
43                     break;
44                 }
45             }
46
47             // If it is a saddle point, print the coordinates
48         }
49     }
50 }
```

